



UNIVERSITY OF LUSAKA

FACULTY OF ECONOMICS, BUSINESS & MANAGEMENT

UNDERGRADUATES PROGRAM

INTRODUCTION TO MACROECONOMICS

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COURSE MODULE

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UNIT: 1

TOPIC: MACROECONOMICS OVERVIEW

OBJECTIVE:

At the end of the lesson, students must understand macroeconomics as the study of aggregates of the whole economy. They should clearly differentiate the micro economic and macro economics concepts and relate the macroeconomic concepts to the economic activities in the world.

SUB- TOPICS:

- Definition of macroeconomics
- Objectives and instruments of macroeconomics

A. Definition of Macroeconomics

- *'Macroeconomics is the study of the behavior of the economy as a whole.'* (Samuelson .N, 2007)
- *'Macroeconomics is the study of aggregate economic variables.'* (Blanchard,O, 2007)
- Examples of aggregates are aggregate production, average price level, unemployment rate, inflation, unemployment, and industrial production. The aggregate will be covered in detail in other lectures.
- The distinction between Microeconomics and Macroeconomics is that they stress on different issues. Macroeconomics considers the industrial sector, the services sector or the farm sector but not consider specific parts of any of these sectors. Microeconomics considers units production and prices in a specific markets such as unit prices of commodities, individual demand for goods and services.

B. Objectives of macroeconomics

- The objective of Macroeconomics is to explain the behavior of the aggregates variables due to the effect of government policies. This is achieved by :
 - Making assumptions to simplify the analysis
 - Construct simple structures (models) to examine and interpret the economy. Examples of models are IS-LM model, AS-AD model, Phillips curve

C. Focus of macroeconomics

The major focus of the macroeconomic is aggregate output, economic growth, unemployment and inflation .

- Aggregate output is quantity of goods and services produced and measured by Gross Domestic Products (GDP) or Gross National Products (GNP).
- Economic growth is the rise in real GDP or GNP and measured by growth of GDP or GNP
- Unemployment is the number of people without jobs out of the labor force and measured by unemployment rate.
- Inflation is the persist rise in relative prices for goods and services and measured by inflation rate

D. Instruments of macroeconomics

- **Macroeconomic instruments** fall within the realm of Macroeconomics policy. The policy is divided into two subsets: Monetary policy and Fiscal policy.
- Monetary policy is conducted by the central bank of a country, and Fiscal policy is conducted by a Government department and deals with managing a nation's Budget. In Zambia, the monetary policy is conducted by Bank of Zambia and Fiscal policy is conducted by Ministry of Finance.
- The policies are used to stabilize the economy and pursue macroeconomic goals.

UNIT: 2

TOPIC: MEASUREMENT OF ECONOMIC PERFORMANCE

LEARNING OBJECTIVE:

At the end of the lesson, students must understand the comprehensive measures of economic performance such as GDP, GNP, National Income and other macroeconomic variables used as economic indicators. They should competently understanding how to compute the National Income account and differentiate the two approaches used in the analysis of National Income: Expenditure and Income approach.

SUB- TOPICS:

- Concepts: GDP,GNP, NI and economic indicators
- Expenditure and Income approach
- National income account computation
- Problems in measuring National Income

1 Concepts: GDP,GNP,NI and other economic indicators

1.1 GDP Concept

- The Gross Domestic Product (GDP) measures the value of all final goods and services produced within a country. The real GDP is when the GDP is adjusted to prices or inflation.
- Composition of GDP is : $C + I + G + (X - M)$, where
- C =consumption, I =Investment, G =Government spending , X =exports , I = imports
- GDP per Capita is the value of goods produced per person in the country. This is equals to the GDP divided by the number of people in the country
- Measuring economic growth rate using GDP:

$$\frac{\text{Real GDP } t - \text{Real GDP } t-1}{\text{Real GDP } t-1} \times 100$$

Where t is the current year and t-1 is the previous year

Illustration

The following information was collected from central Intelligence Agency (CIA) fact book website with the IMF GDP, population estimates, inflation and unemployment for the member countries. Using the information in the table, calculate the economic growth and GDP per capita for Zambia,

Table 1: GDP, Population, Inflation ,Unemployment ,Labor force Estimates

	GDP (\$million)		Population	Inflation		Unemployment (million)		Labour force (million)
Country/Year	2009	2008	2009	2009	2008	2009	2008	2009
Zambia	14,958.13	14,314.00	12,056,923	14%	12.5%	2699	49%	5398

Source: CIA Factbook, 2009

Economic growth Solution:

$$\text{Rate of GDP} = \frac{14,958,130,000 - 14,314,000,000}{14,314,000,000} \times 100 = 4.5\%$$

Analysis: The 2009 economic growth for Zambia was 4,5 %

Table 2: Actual and Projected Zambia Economic Growth Rates

Year	2007	2008	2009	2010
Projected	7	7	5	5.5
Actual	6.2	6	4.5	5

Note: Estimate Government Projections

Source: MoFNP (2008, 2009)

- Analysis for the variance in the actual and projected economic growth can be sourced from The Global Financial Crisis Discussion series (Professor .M. Ndulo, Mudenda.D, Ingombe.L, Muchimba.L (May 2009). www.odi.uk/resources/downloads

Exercise: Refer to table 1 Calculate the GDP per capita for 2009

1.2 GNP Concept

- Gross National Product (GNP) measures the value of all final goods and services produced with domestically owned factors of production. The concept is now commonly referred to as Gross National Income.
- Measuring economic growth rate using GNP similar to the GDP approach:

$$\frac{\text{Real GNP } t - \text{Real GNP } t-1}{\text{Real GNP } t-1} \times 100$$

Where t is the current year and t-1 is the previous year

- The difference between the GDP and GNP is that GDP measures the value of all final goods and services produced **within** a country **less** payment to foreign factors of production, **plus** factor income of residents abroad. While GNP measures the value of all final goods and services produced with **domestically owned** factors of production.

1.3 NI

- National income measures how much the economy can spend (or save) after setting aside resources to maintain the capital stock by offsetting depreciation. NI is also referred to as Net National Product (NNP).
- National income (Net National Product) is calculated by :

GNP- depreciation

1.4 Other Economic Variables

Other variables are used in the analysis of the economic performance as economic indicators such as Inflation rate and Unemployment rate.

1.3.1 Inflation rate is the percentage change in the average price of goods and services and calculated as follows:

$$\frac{\text{CPI } t - \text{CPI } t-1}{\text{CPI } t-1} \times 100$$

Where t is the current year, t-1 is the previous year and CPI is consumer price index

- The Consumer Price Index (CPI) is a measure of the average change of prices over time for a constant goods and services basket purchased by a representative consumer group. The deflator is used in the absence of CPI data. The deflator is the measure of the average change of prices over time for a fixed goods and services basket produced in the economy.
- Other various indexes have been devised to measure different aspects of inflation such as the Producer Price Index (PPI) that measures inflation at earlier stages of the production process; the Employment Cost Index (ECI) that measures it in the labor market; the BLS International Price Program that measures it for imports and exports. The CPI is generally the best measure for inflation.

Table 2: CPI vs Deflator

CPI	Deflator
- Goods basket purchased by a representative group.	- All produced goods in the economy
- Imported consumer goods included	- Imported goods not included
- Common basket	- Variable basket
- Base year: Changes in commodity basket	- Base year problems: New products
- Goods and quality change	- Quality change

Exercise: Refer to table 1. Calculate the 2009 inflation rate for Zambia

Illustration : Calculation of CPI

How to Calculate the Consumer Price Index .Consider a very simple example:

Anna's market basket is made up of four textbooks and two cups of coffee. The prices of those goods are as follows:

<u>Year</u>	<u>Price of Textbooks</u>	<u>Price of Coffee</u>
2000	\$1	\$2
2001	\$2	\$3
2002	\$3	\$4

Now that we know the contents of the market basket and the price of the relevant goods in the years of interest, the next step is calculate the cost of the market basket for each year.

<u>Year</u>	<u>Cost of Market Basket</u>
2000	$(4 \times \$1) + (2 \times \$2) = \$8$
2001	$(4 \times \$2) + (2 \times \$3) = \$14$
2002	$(4 \times \$3) + (2 \times \$4) = \$20$

The last step is to use the market-basket costs to calculate the CPI (assuming 2000 is the base year):

<u>Year</u>	<u>Consumer Price Index</u>
2000	$(\text{Cost}_{00} / \text{Cost}_{00}) \times 100 = \$8 / \$8 \times 100 = 100$
2001	$(\text{Cost}_{01} / \text{Cost}_{00}) \times 100 = \$14 / \$8 \times 100 = 175$
2002	$(\text{Cost}_{02} / \text{Cost}_{00}) \times 100 = \$20 / \$8 \times 100 = 250$

1.3.2 Unemployment rate is the percentage fraction of the labor force without a job and is calculated as follows:

$$\frac{\text{Number of Unemployed}}{\text{Total Labor force}} \times 100$$

- A high unemployment rate implies reduction in the number of people employed out of the labor force and low unemployment rate implies an increase in the number of people employed.

Exercise: Refer to table 1. Calculate the 2009 unemployment rate for Zambia

2 Expenditure and Income Approach

Income and expenditure accounts provide two of the three ways to calculate gross domestic product (GDP), the income approach and the expenditure approach (the third way being the value added approach).

2.1 Income Approach

- Income-based GDP is calculated by adding earnings from the factors of production labor (wages and salaries), capital (rent and interest), production (profits and taxes) less subsidies, in order to obtain a measure comparable to that of expenditure-based GDP.

Computation:

Table 3: Income Based GDP Mathematical Example

Add		\$ Millions
+	Wages, salaries and supplementary labour income	650,000
+	Corporation profits before taxes	180,000
+	Government business enterprise profits before taxes	10,000
+	Interest and miscellaneous investment income	60,000
+	Accrued net income of farm operators from farm production	2,000
+	Net income of non-farm unincorporated business, including rent	80,000
+	Inventory valuation adjustment	1,000
+	Taxes less subsidies	150,000
+	Capital consumption allowances	175,000
+	Statistical discrepancy	500
=	GDP at Market prices	1,308,500

Source: Canada department of statistics, 2009

2.2 Expenditure Approach

- Expenditure-based GDP is calculated by adding the final expenditures of the four sectors of the economy: persons, businesses, governments, and non-residents (import and exports).

Table 3: Expenditure Based GDP Mathematical Example

Add		\$ Millions
+	Personal expenditure on consumer goods and services	710,000
+	Government current expenditure on goods and services	250,000
+	Government gross fixed capital formation	35,000
+	Government investment in inventories	20
+	Business gross fixed capital formation	225,000
+	Business investment in inventories	8,980
+	Exports of goods and services	500,000
-	Deduct: Imports of goods and services	420,000
-	Statistical discrepancy	500
=	GDP at Market prices	1,308,500

Source: Canada department of statistics, 2009

2.3 National Income Account Computation

- The national Income account is constructed from business accounts .In the business income statement the left hand side shows the sales of Goods (output) and right hand side shows the cost of production and (wages rents which are earnings).The National account then adds up the aggregates of number of identical firms to obtain total GDP (See page 427 ,Samuelson ,2007).
- Note that the GPD the Expenditure (product) side is equal to the GDP from the Income (earnings) side.
Refer to table 3 and 4.

3 Problem in Measuring National Income Using GDP

The famous French economist Alfred Sauvy, points to the fact that GDP excludes (or significantly underestimates) goods and services produced outside the official market economy. GDP deficiencies can be divided into four categories:

- Non-market production (household activities) - domestic work and business such tailoring, home based salons and barbers shops are not accounted for in GDP .There is no way to measure the value added by these service
- Government services (valuation problems)- In public accounting rules, GDP is the sum of values added by all economic entities, Thus, value added is in fact constituted by two main parts: wages and profits. In Government ministries, no value of production is available as public services are generally not bought by anyone on a market (think of public gardens maintenance or tax collection). A free service resulting from a past public investment does not appear in GDP
- Informal sector (underground economy) - According to various studies carried out in Zambia, domestic production could represent as much as 60% of the national outputs
- Voluntary work: a bicycle repaired by a friend makes GDP fall if the work used to be done by (and paid to) a professional. Thus, a society where voluntary work is widespread will enjoy a higher level of economic well-being but not necessarily a higher GDP (in fact, even a lower GDP
- Other deficiencies
 - GDP and economic well-being – GDP only considers the positive side of production and not the negative externalities of production. These contribute to environmental degradation, pollution, poor health and subsequently an increase economic cost.
 - GDP and non-economic well-being (welfare) – There are numerous technical difficulties in the measurement of distribution of income and wealth, life expectancy and health, education and access to education that possess varies in Human Development Index (HDI) for developing and developed countries.

TOPIC: NATIONAL INCOME

LEARNING OBJECTIVE:

At the end of the lesson, students must understand the components of demand at national level and apply the expenditure and multiplier model to explain national Income equilibrium. Finally they should understand the relationship between savings and investments decisions.

SUB- TOPICS:

- Aggregate demand and output
- Aggregate Expenditure model
- National Income equilibrium
- The multiplier model
- Paradox of shifts -Savings & Investments

1 Aggregate demand and output

1.1 Definition and determinants of AD

1.1.1 Definitions

- Aggregate demand (AD) is the total demand for final goods and services in the economy (Y) at a given time and price level. It is the amount of goods and services in the economy that will be purchased at all possible price levels – Keynes
- The total amount that different sectors in the economy are willingly spend in a given period and it depends on levels of prices - Samuelson

1.1.2 Determinants of AD

- We assume that what is produced (Y) is equal to what is demanded (AD)
- In a closed economy without a government and foreign sector, sources of AD are Consumption (C) and Investments (I)

$$AD = C + I$$

- In a closed economy with a government, sources of AD are Consumption (C) Investments (I) and government spending (G)

$$AD = C + I + G$$

- Open economy – according to Mundell- Fleming, there are four major components of aggregate demand. The equation for aggregate demand is $Y = C(Y - T) + I(r) + G + NX(e)$, where

Y represents income or output.

$C(Y - T)$ represents consumption as a function of disposable income, defined as income Less tax

$I(r)$ represent investment as a function of the interest rate,

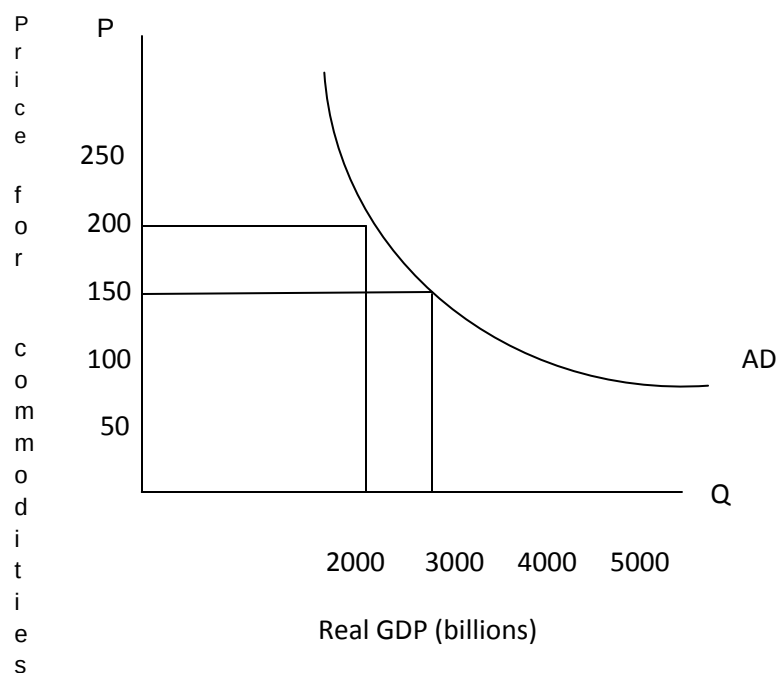
G represents government spending,

$NX(e)$ represents net exports, defined as exports less imports (X-IM)

1.1.3 Downward sloping AD Curves

- The aggregate demand curve is just like any other demand curve, but for the sum total of all goods and services in an economy. The aggregate demand curve can be thought of just like a demand curve for a firm. When the price level is high, aggregate demand is low; when the price level is low, aggregate demand is high.
- It shows a downward slope with price level on the vertical axis and income or output on the horizontal axis. It lies in a plane consisting of the price level and income or output.
- The aggregate demand curve outlines the relationship between income or output and the price level. It is important to notice that aggregate demand is a schedule because as the price level changes, the income or output also changes.

Figure 1: AD Curve



Source: Samuelson, 2007

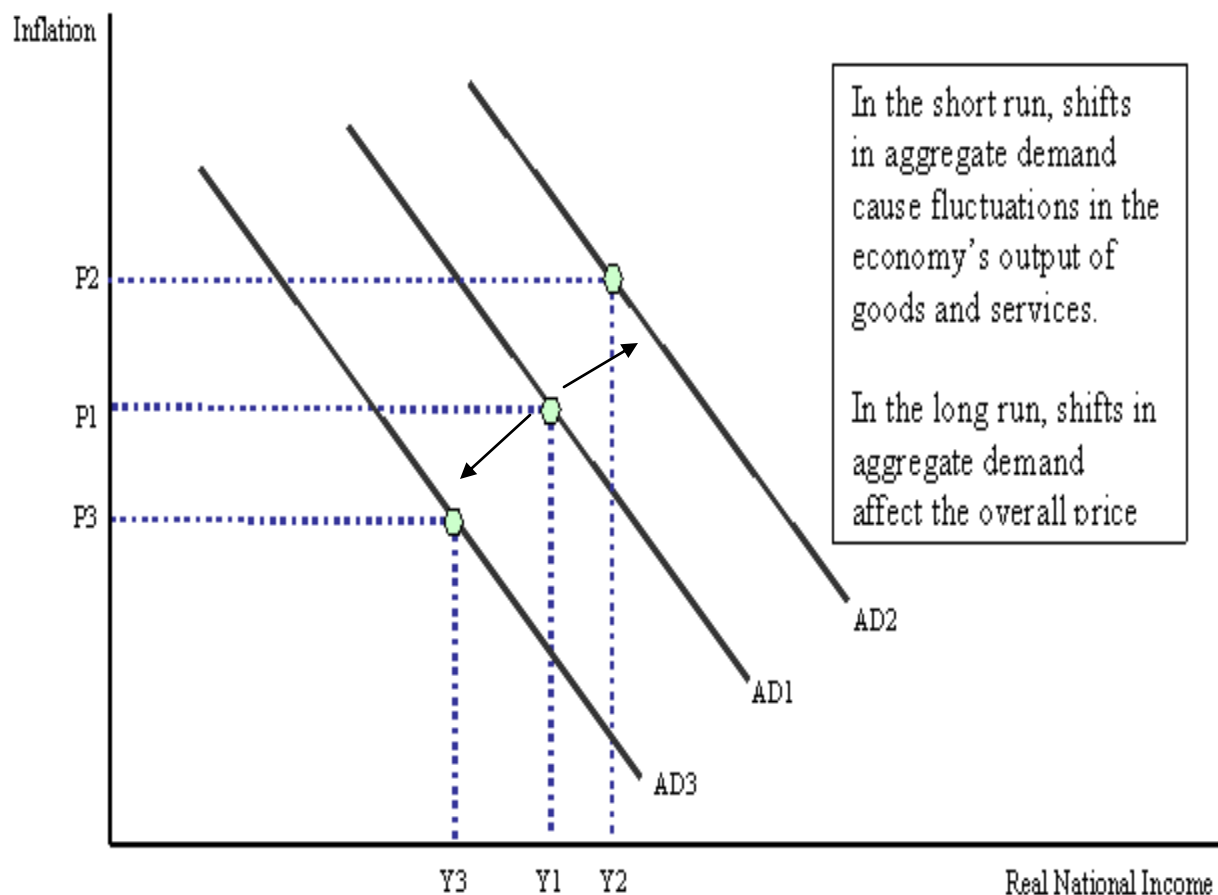
- The AD curve represents the quantity of total spending at different price levels, with other factors held constant.
- The aggregate demand curve is downward sloping. There are a number of reasons for this relationship. A downward sloping aggregate demand curve means that as the price level drops, the quantity of output demanded increases. Similarly, as the price level drops, the national income increases.
- The equation for aggregate demand of $Y = C(Y - T) + I(r) + G + NX(e)$ has been deciphered. This equation has many meanings such as output, national income, and GDP.

Exercise: Why is the AD downward sloping?

1.1.4 Shifts in the AD Curves

A change in factors affecting any one or more components of aggregate demand, households (C), firms (I), the government (G) or overseas consumers and business (X) changes planned aggregate demand and results in a shift in the AD curve

Figure 2: Shifts in the AD Curve



Source: Geoff Riley, 2006

- The diagram above which shows an inward shift of AD from AD1 to AD3 and an outward shift of AD from AD1 to AD2. The increase in AD might have been caused by a fall in determinant of AD or an increase in the determinant of AD.

Table 4: Summary of the causes in the AD Curve shifts

Factors causing a shift in AD	
Changes in Expectations Current spending is affected by anticipated future income, profit, and inflation	- <u>The expectations</u> of consumers and businesses can have a powerful effect on planned spending in the economy E.g. expected increases in consumer incomes, wealth or company profits encourage households and firms to spend more – boosting AD. Similarly, higher expected inflation encourages spending now before price increases come into effect - a short term boost to AD. - When confidence turns lower, we expect to see an increase in saving and some companies deciding to postpone capital investment projects because of worries over a lack of demand and a fall in the expected rate of profit on investments.
Changes in Monetary Policy – i.e. a change in interest rates (Note there is more than one interest rate in the economy, although borrowing and savings rates tend to move in the same direction)	- An expansionary monetary policy will cause an outward shift of the AD curve. If interest rates fall – this lowers the cost of borrowing and the incentive to save, thereby encouraging consumption. Lower interest rates encourage firms to borrow and invest. - There are time lags between changes in interest rates and the changes on the components of aggregate demand.
Changes in Fiscal Policy Fiscal Policy refers to changes in government spending, welfare benefits and taxation, and the amount that the government borrows	- For example, the Government may increase its expenditure e.g. financed by a higher budget deficit, - this directly increases AD - Income tax affects disposable income e.g. lower rates of income tax raise disposable income and should boost consumption. - An increase in transfer payments rises AD – particularly if welfare recipients spend a high % of the benefits they receive
Economic events in the international economy International factors such as the exchange rate and foreign income (e.g. the economic cycle in other countries)	- A fall in the value of the kwacha (a depreciation) makes imports dearer and exports cheaper thereby discouraging imports and encouraging exports – the net result should be that ZMK AD rises – the impact depends on the price elasticity of demand for imports and exports and also the elasticity of supply of ZMK exporters in response to an exchange rate depreciation. - An increase in overseas incomes raises demand for exports and therefore ZMK AD rises. In contrast a recession in a major export market will lead to a fall in ZMK exports and an inward shift of aggregate demand. The ZMK is an open economy, meaning that a large and rising share of our national output is linked to exports of goods and services or is open to competition from imports.
Changes in household wealth Wealth refers to the value of assets owned by consumers e.g. houses and shares	- A rise in house prices or the value of shares increases consumers' wealth and allow an increase in borrowing to finance consumption increasing AD. In contrast, a fall in the value of share prices will lead to a decline in household financial wealth and a fall in consumer demand.

2 Aggregate Expenditure Model

2.2.1 Definition

- Aggregate expenditure (**AE**) is the sum of consumption, investment, government purchases, and net export. Of these four sectors, the consumption represents the largest share.

2.2.2 The consumption function

- The consumption function shows the positive relationship between disposable income and consumption demand. It shows aggregate consumption demand at each level of personal disposable income

$$C = C_o + MPC (Y_d), \text{ where :}$$

C = total consumption

C_o = autonomous consumption demand whose amount is independent of disposable income

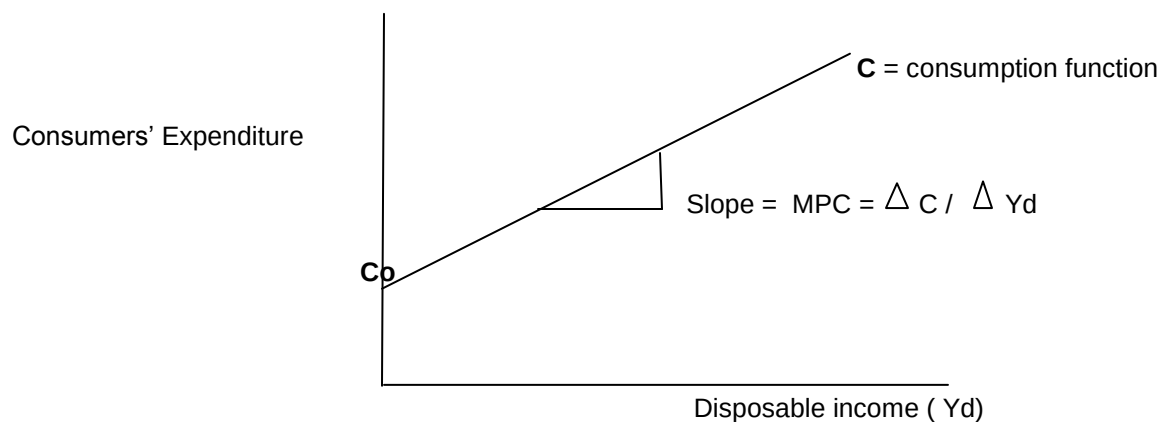
MPC = marginal propensity to consume is a fraction of each extra pound of disposable income that household wish to

Consume, The remaining (1- 0) they wish to save . This is a fraction between 0 and 1; It is equal to change in consumption brought about by a change in disposable income. It is the slope of the consumption function.

$$MPC = \text{change in } C / \text{change in } Y_d \text{) , where}$$

Y_d = disposable income.

Figure 3: Consumption Function



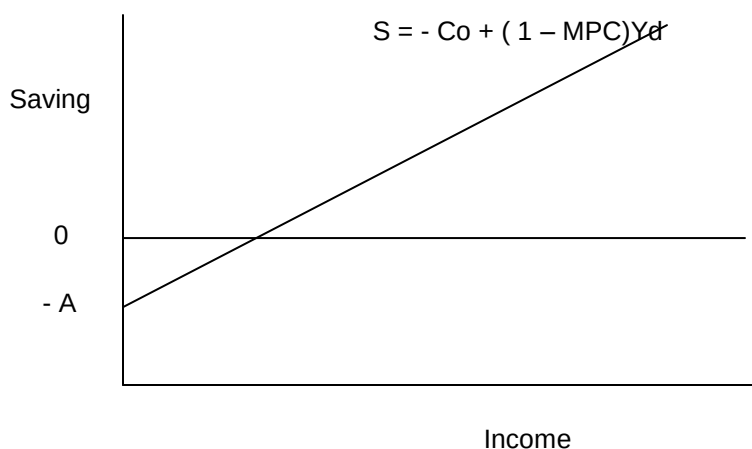
Source: Begg.D,2007

2.2.3 The Saving function

- The saving function shows the desired saving at each income level. Saving is income not consumed. The saving function is derived from the consumption function :

$$S = -C_0 + (1 - MPC) Y_d$$

Figure 4 : Saving Function



Source: Begg. D, 2007

- MPS = marginal propensity to save is equal to change in savings (S) brought about by a change in disposable income.

$$MPS = \text{change in } S / \text{change in } Y_d$$

$$MPS = 1 - MPC$$

- Since all income must be either consumed or saved, then any change in income must also be consumed or saved.

$$\text{Therefore: } MPC + MPS = 1$$

- The average propensity to consume (APC) is the portion of income spent on consumption.

$$(APC = C / Y_d)$$

- The average propensity to save (APS) is the portion of income saved.

$$(APS = S / Y_d)$$

$$\text{Again, } APC + APS = 1$$

- Therefore $S + C = Y_d$, planned saving is part of income not planned to be spent on consumption

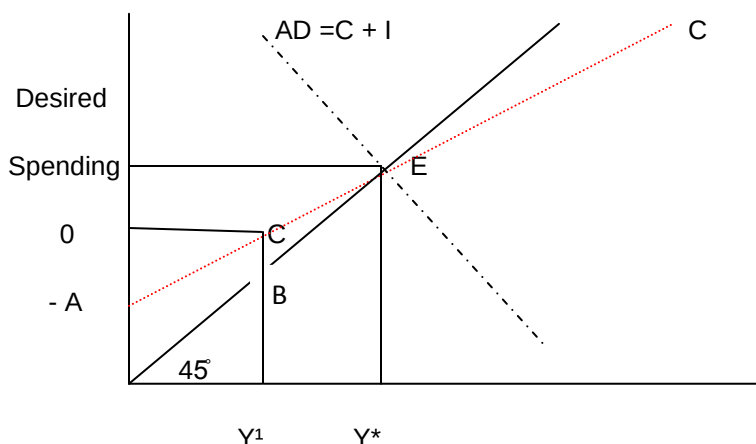
3 National Income Equilibrium

3.1 Definition

- Equilibrium is when planned saving equals actual output and actual income – Begg, D

Equilibrium of National Income(Y) is the level of output whose production will create total spending just sufficient to purchase that output. If the economy produces an amount of goods that differs to the amount that the four sectors of the economy buy (AE), AE and aggregate production (AP) are not equal; then the economy is in disequilibrium.

Figure 5: The 45° Line diagram and equilibrium Output



Source: Begg, 2007

- The 45° line reflects any value on the horizontal axis onto the same value of the vertical axis. Point E, where AD schedule crosses the 45° line is the point where AD is equal to income. Therefore E is the equilibrium point at which planned spending equals actual output and actual income.
- All output below E are Y^* , aggregate demand exceeds income and output and output above E implies that AD is less than income and output.

Adjustments towards equilibrium

- When output (Y) is above E output (Y^*) this implies that $AE < AP$, firms will involuntarily accumulate inventory. This will signal firms that they have overproduced. As a result, firms will cut back on production and/or prices. This will decrease the total value of output, moving the economy towards the equilibrium GDP.
- When output (Y^1) is below E output (Y^*) this implies that $AE > AP$, inventories will be depleted unexpectedly. This will signal firms that they have not produced enough. As a result, firms will increase productions and/or prices. This will increase the total value of output, moving the economy towards the equilibrium GDP.

4 The Multiplier Model

- The multiplier model is the ratio of the change in equilibrium output to the change in autonomous spending that caused the change (Begg , 2007).It tells how much output has changed after a shift in AD. Recall that $AD = C + I$. In this analysis we will focus on a parallel shift in the AD schedule caused by a change in autonomous investments demand.
- The multiplier depends on the MPC; Keynes observed that changes in autonomous expenditures (those expenditures independent of income) could create even larger changes in national income. Therefore, a large change in MPC , the multiplier is small or vice versa.

$$\text{Multiplier} = 1 / (1 - \text{MPC}) = 1/\text{MPS}$$

- For example, a technological breakthrough has increased the autonomous investment by \$50million, this \$50M will become the income of the resources market. Workers who may have higher income are willing to spend more in the market. Their spending becomes the income of producers who will again spend in the market, and create extra income. This process repeats itself, creating a multiplier effect. If the Multiplier (M) = 2.5, then the aggregate expenditure will increase by $\$50\text{M} \times 2.5 = \125M .

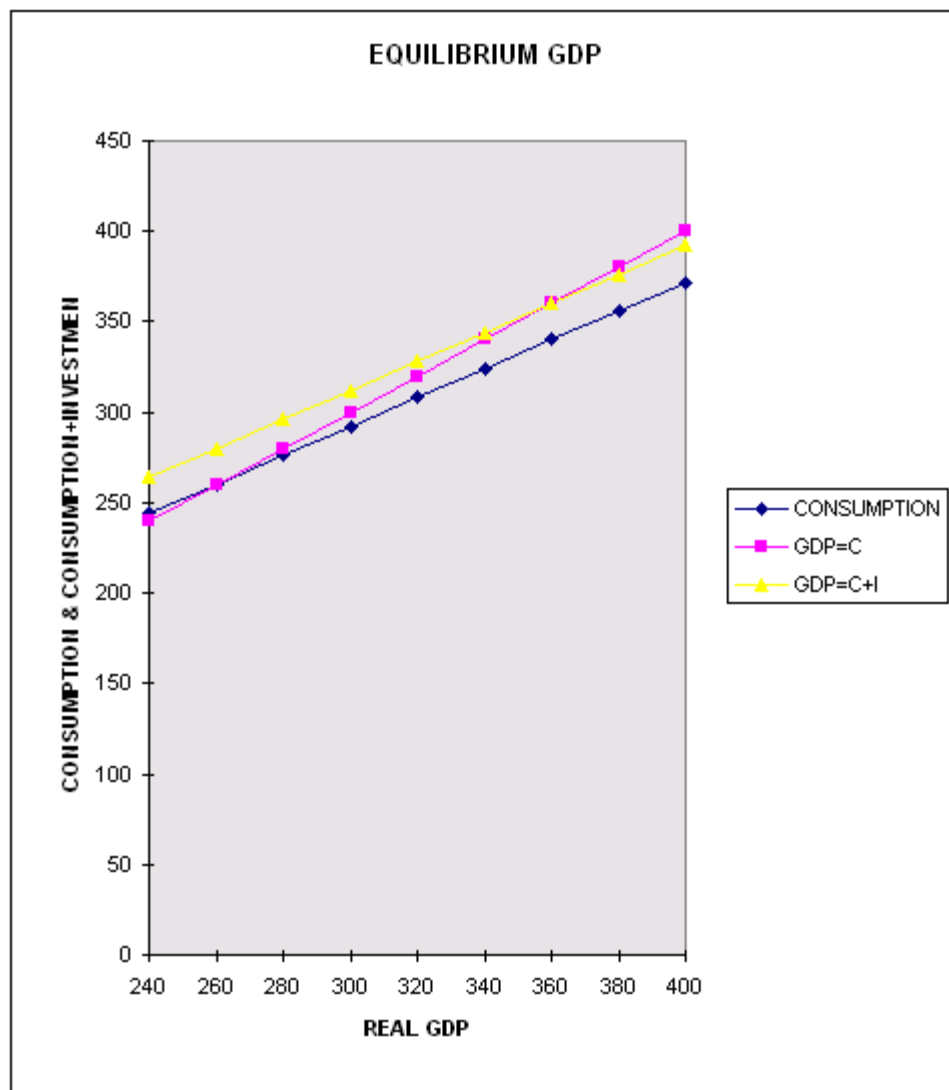
4.1 Equilibrium using the Multiplier Models

The equilibrium between AE and AP and the multiplier model is illustrated below.

Table 5: Data in this table is used to construct the graph below.

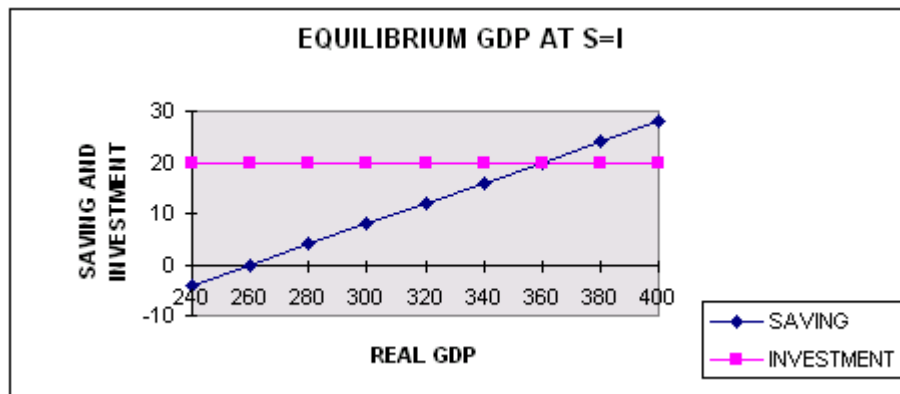
GDP	CONSUMPTION	GDP=C+I	SAVING	INVESTMENT
240	244	264	-4	20
260	260	280	0	20
280	276	296	4	20
300	292	312	8	20
320	308	328	12	20
340	324	344	16	20
360	340	360	20	20
380	356	376	24	20
400	372	392	28	20

Figure 6: The equilibrium between AE and AP (AE & Y)



- The investments has increased the equilibrium GDP, (from the intersection of red and blue curves (A) to the new intersection of red and yellow curves (B).The same equilibrium GDP can be obtained by observing the Investment and Saving schedule.

Figure 7: Equilibrium using Investments and saving Model and multiplier Model



- In both graphs, we can see that equilibrium GDP is at 360.
- Another approach to get the equilibrium GDP is by using the multiplier. The equilibrium GDP, $GDPe$, before the investment is 260. MPC for this data set is 0.8.

$$\text{Multiplier} = 1 / 1 - \text{MPC} = 1 / (1 - 0.8) = 5$$

The investment amount of 20, will increase $GDPe$ by 5 times, this will give us an increase of

$20 \times 5 = 100$. Therefore, the new equilibrium GDP will be $260 + 100 = 360$.

5 The Paradox of thrift

- Paradox of thrift is the change in the amount households wish to save of each income leads to a change in E income, but no change in equilibrium saving which must be equal planned investments (Begg, 2007). In this analysis, we will focus on a parallel shift in the AD schedule caused by a change in the autonomous consumption and savings.
- A higher autonomous consumption demand implies an identical fall in autonomous planned saving. For example, if autonomous consumption increases by 5, there is a parallel shift in the consumption function and AD but a parallel downward shift in the saving function.
- In Equilibrium: Planned S = Planned I (investment is not altered).

Therefore at E, income must always adjust to restore planned savings to the unchanged level of planned investments.

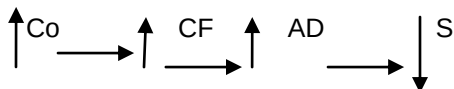
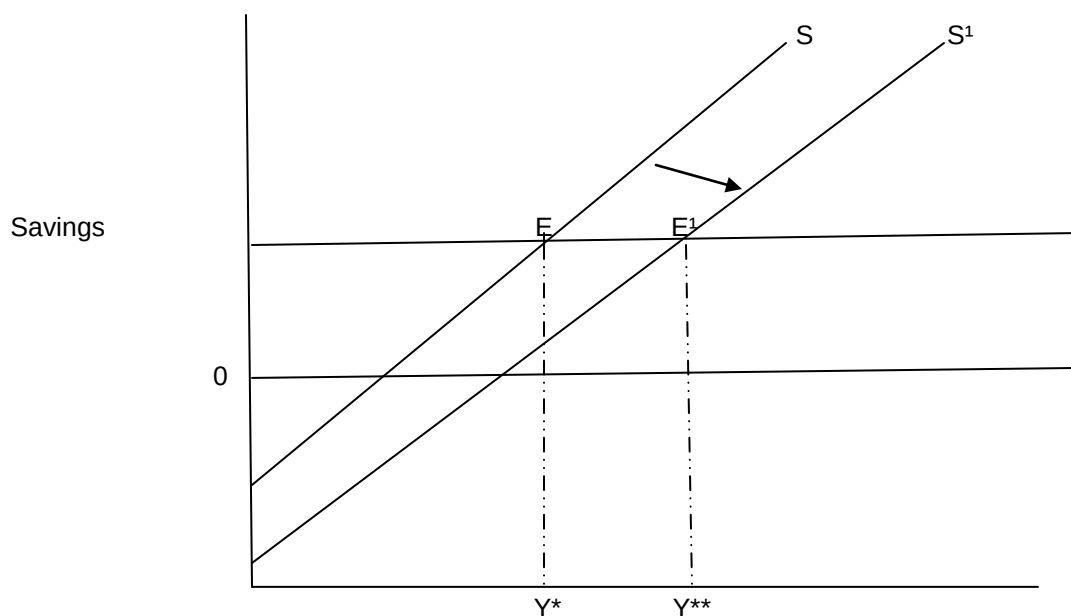


Figure 8: The paradox of thrift (Savings & Investments)



Source: Begg, D, 2007

- In equilibrium, planned savings equals planned investments. A fall in the desire to save induces a rise in output to keep planned saving equal to planned investments.

UNIT: 4

TOPIC:

FISCAL POLICY

LEARNING OBJECTIVE:

- ❑ Students must understand the instruments used by the Government under fiscal policy in stabilizing the economy.
- ❑ Students must apply the applicable fiscal policy in an economic situation to stabilize the economy.

SUB- TOPICS:

- Goals of fiscal policy
- Tools of Fiscal policy
 - Government spending
 - Taxes
- Fiscal Policy Expansion and Contraction
- Effects of Governments spending and taxes on output
- Crowding out effect
- Fiscal Policy and budget deficit:
 - Government and Structural budget
 - Budget deficits adjustments
- The national debt and the deficit
- Limitations of Fiscal policy

1. GOALS OF FISCAL POLICY

- Fiscal policy is one of the instruments that government uses to stabilize the economy. The policy can be used to reduce inflation, unemployment, improve economic growth or correct trade balance. Fiscal policy consists of managing the national Budget and its financing so as to influence economic activity (Samuelson. N, 2006)
- This entails the expansion or contraction of government expenditures related to specific government programs. It also includes the raising of taxes to finance government expenditures and the raising of debt (bonds & treasury bills) to bridge the gap (Budget deficit) between revenues (tax receipt) and expenditures related to the implementation of government programs.

1.1 Tools of fiscal Policy

- The tools of Fiscal policy are taxes and government spending:
 - Government spending entails government expenditures related to specific government programs
 - Taxes is the raising or reduction of taxes to finance government expenditures

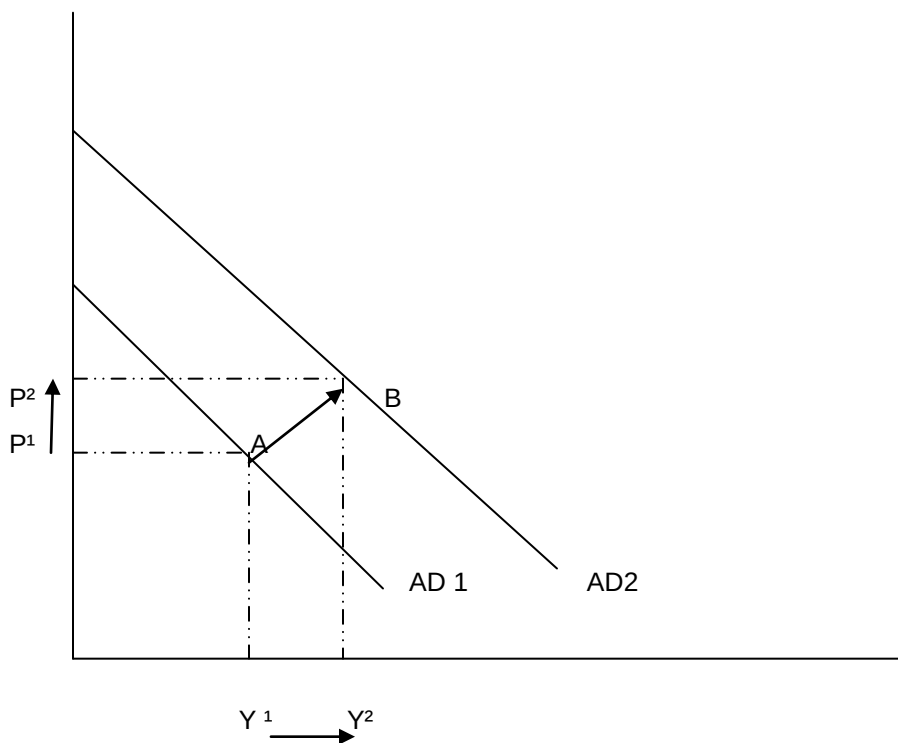
- Government spending affects the overall spending in the economy, while taxes affect individuals disposable income and prices of goods and factors of production.

2. Fiscal Policy Expansion and Contraction

2.1 Fiscal Policy Expansion

- Fiscal policy expansion is when government spending is increased to increase output. Taxes are reduced to increase disposable income to increase aggregate demand (AD) and output. Fiscal policy expansion is adopted during recession to foster economic growth.
- Lowering taxes and increasing the Budget Deficit is considered an expansive fiscal policy that would increase aggregate demand and stimulate the economy.

Figure 1: Fiscal policy expansion



Source: Blanchard. O, 2007

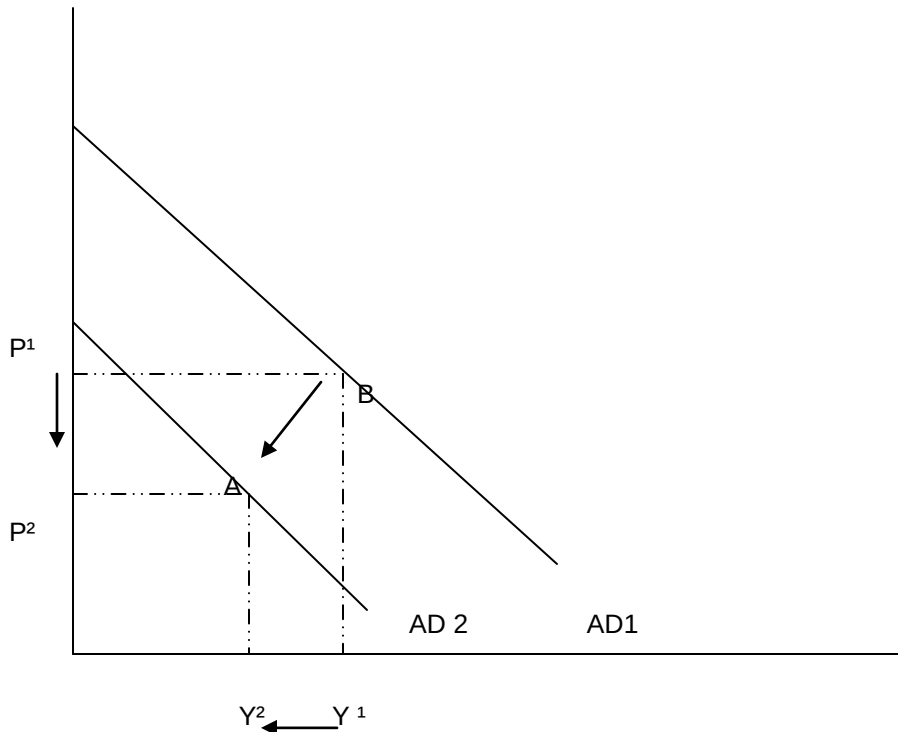
When government spending is increased or taxes are reduced AD increases from AD1 to AD 2, output increases from Y^1 to Y^2 and general prices increase from P^1 to P^2 . Therefore overall economic activity increases when fiscal policy expansion is implemented.

2.2 Fiscal Policy Contraction

Fiscal policy contraction is when government spending is reduced to decrease output. Taxes are increased to reduce disposable income and aggregate demand (AD). Subsequently output is reduced. Fiscal policy contraction is adopted during a boom to slow down economic growth.

- Raising taxes and reducing the Budget Deficit is deemed to be a restrictive fiscal policy as it would reduce aggregate demand and slow down GDP growth.

Figure 2: Fiscal policy contraction



Source: Blanchard. O, 2007

When government spending is reduced or taxes are increased AD reduces from AD_1 to AD_2 , output reduces from Y_1 to Y_2 and general prices reduce from P_1 to P_2 . Therefore overall economic activity slows down when fiscal policy contraction is implemented.

3. EFFECTS OF GOVERNMENT SPENDING AND TAXES ON OUTPUT

3.1 Under fiscal policy expansion

- Increase in government spending leads to an increase in output. This eventually leads to an increase in employment. Therefore AD and prices also rise.

- Reduction in taxes leads to an increase in the disposable income of individuals. This increases the AD and stimulates economic activities (output increases).

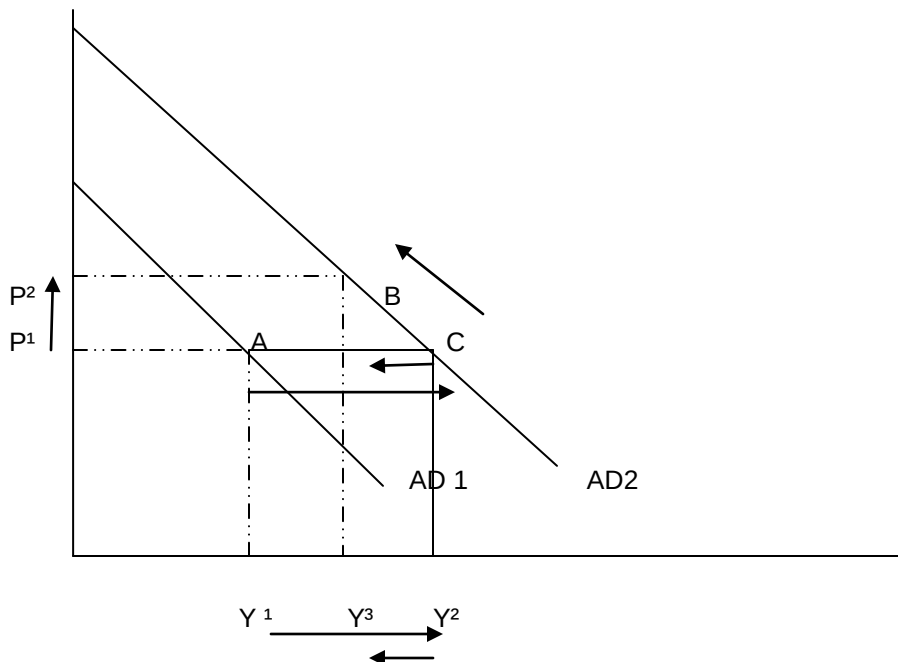
3.2 Under Fiscal Contraction

- Reduction in government spending leads to a decrease in output. This eventually leads to decrease in employment, AD and prices also fall down.
- Increase in taxes leads to a reduction in the disposable income of individuals. This decreases the AD and slows down economic activities (output reduces).

4. CROWDING OUT EFFECT

- As government spend more to correct the recessionary gap in the economy, they compete with private sector for resources and goods and services.
- As government expenditure increases, consumption and investment decreases, causing the ineffectiveness of the fiscal policy.
- Another way of defining crowding out effect is when government increases spending and reduces taxation, resulting in a deficit which must be funded through borrowing. This increased borrowing will push the interest rate higher, reducing the consumption and investment in the private sector.

Figure 3: Crowding out effect



Source: Blanchard. O, 2007

- At the same level of prices, an increase in government spending will lead to an immediate shift of AD from point A to Point C at AD₂. Output will automatically increase from Y^1 to Y^2 . At point C, government will start competing with private sector for resources and goods and services. Therefore part of the investment sector will fall out of business and output will reduce from Y^2 to Y^3 . This will also lead to a fall in AD₂ along the same demand curve from point C to B. The fall of output from Y^2 to Y^3 and aggregate demand from point C to B is referred to as crowding out effect.

5. **FISCAL POLICY AND BUDGET DEFICIT**

- Budget deficit is a shortfall between Government revenue – expenditure. This is when government expenditure is less than the revenue. The source of government revenue is taxes.
- Therefore raising of taxes to finance government expenditure is a good measure to reduce budget deficit (Fiscal contraction).
- Lowering taxes and increasing government spending is considered an expansive fiscal policy that would increase aggregate demand and stimulate the economy. However, this leads to an increase in the Budget Deficit.

5.1 **Budget deficits adjustments**

- The budget deficit should be adjusted to the structural budget level. The structural budget shows a balanced budget where the government expenditure is equal to revenue (what the budget would be like if output were at potential output.)
- Given that higher government spending on goods and services increases equilibrium. Given a tax rate, revenue rises but the deficit increases. (Note that budget deficit may not be necessarily be connected to fiscal stance. A deficit may occur during a recession due to low output and the deficit is larger than in the boom.)
- Budget deficit corrective measure is to impose a higher tax rate that will reduce disposable income and affect costs of factors of production. This will lead to a reduction in AD and output taking the equilibrium output to the equilibrium output point (structural budget). Therefore the budget deficit is reduced.
- Alternatively government spending can be reduced which is a very rare measure taken to correct budget deficit.

6. THE NATIONAL DEBT AND THE BUDGET DEFICIT

- When government implements an expansionary fiscal policy, Government increases spending and reduces taxation. This results in a budget deficit which must be funded through borrowing since taxes are reduced.
- Therefore government resorts to borrowing and national debt is created.

7. LIMITATIONS OF FISCAL POLICY

- Lowering taxes and increasing government spending is considered an expansive fiscal policy that would increase aggregate demand and stimulate the economy. This is a restrictive fiscal policy_as it would reduce aggregate demand and slow down GDP growth despite increasing government revenue. This is referred to as crowding out effect.
- Lowering taxes and increasing government spending is considered an expansive fiscal policy that would increase aggregate demand and stimulate the economy. However, this leads to an increase in the Budget Deficit.
- Budget deficit which must be funded through borrowing since taxes are reduced leads to an increase in national debt.

TOPIC: MONETARY POLICY

LEARNING OBJECTIVE:

- ❑ Students must understand the instruments used by the Government under monetary policy in stabilizing the economy.
- ❑ Students must apply the applicable monetary policy in an economic situation to stabilise the economy.

SUB- TOPICS:

- Goals of monetary policy
- Tools of monetary policy
 - Discount rate
 - Reserve ratio
 - Open markets operations
- Monetary policy expansion and contraction (IS –LM Model)
 - Monetary policy in the open market

1. GOALS OF MONETARY POLICY

- Monetary policy is one of the instruments that government uses to stabilize the economy. The policy can be used to manage or curb domestic inflation and control money supply. Monetary policy consists of managing discount rates, reserve requirement for commercial banks and open market operations in the economy (Samuelson. N, 2006).
- Monetary policy is implemented by the central bank of every country (Bank of Zambia in Zambia). Recently, the Central banks have often focused on a second objective, that is managing economic growth as both inflation and economic growth are highly interrelated.

2. Tools of Monetary Policy

- The tools of monetary policy are discount rates, statutory reserve ratios and open market operations.
 - Discount rate is interest rate overnight cash rate for lending money to commercial banks by the central bank. The central bank controls the monetary base by either reducing or increasing the discount rate when lending money to banks.
 - Statutory reserve ratio is the statutory minimum amount every commercial bank is required to have at the central bank. The central bank controls the monetary base by either reducing or increasing the required reserve ratio.

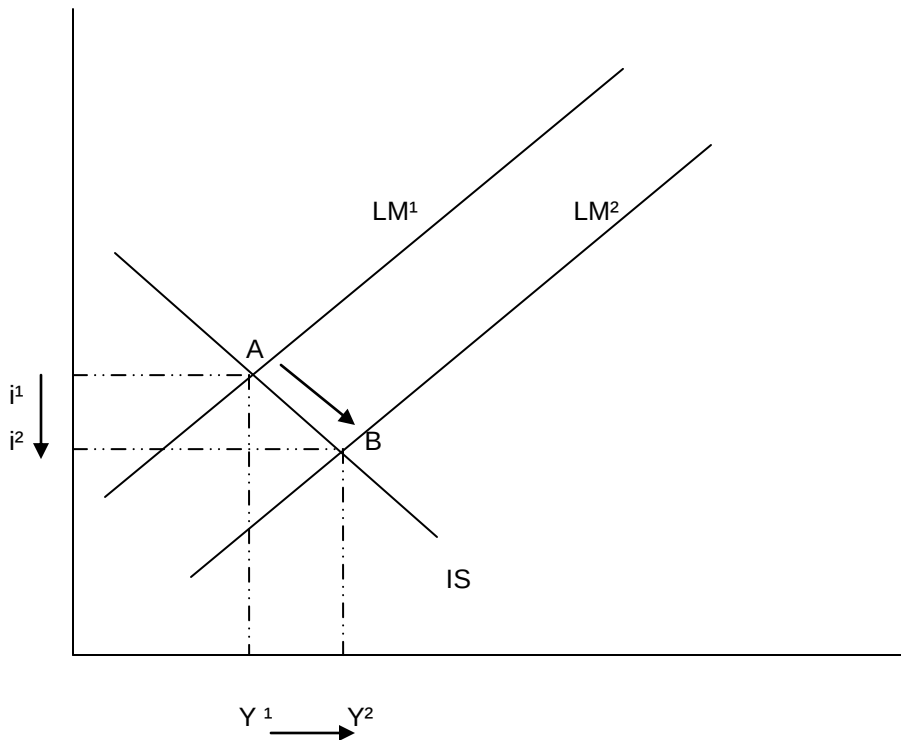
- The open market operations is the selling of treasury bill and bonds to the market with interest to the public. Treasury bills are short term with a maturity period of within one year. Bonds are long term with maturity period after one year. The central bank controls the money in circulation by either selling or buying bonds and treasury bills.

3. Monetary Policy Expansion and Contraction

3.1 Monetary Policy Expansion

- Monetary policy expansion is when
 - Discount rate is reduced to increase money supply in circulation. The cost of borrowing for commercial banks from the central bank is reduced leading to an increase in the money borrowed. The reduced discount rates trickle down to low interest rates charged by the commercial banks to the public. This leads to an increment in borrowing from the commercial and Investment in the economy increases.
 - Reserve ratios are reduced to increase money supply in circulation .The commercial banks monetary base will increase .Therefore money to given to the public will increase.
 - Bonds and treasury bills are bought from the public in exchange for money .This leads to an increase in money in circulation and fosters economic growth.
- Monetary policy expansion is adopted during recession to foster economic growth.

Figure 1: Monetary policy expansion



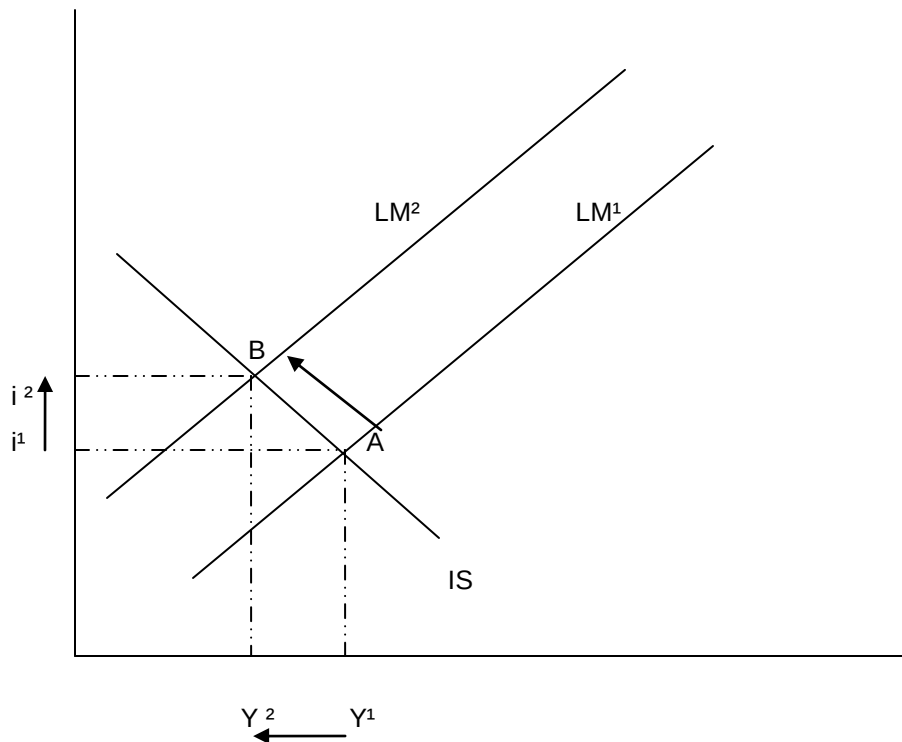
Source: Blanchard. O, 2007

At the same output level (Y^1) interest rate is reduced from i^1 to i^2 . This leads to an increase in money supply, LM^1 shifts downwards to the right to LM^2 . Output increases from Y^1 to Y^2 . The equilibrium point shifts from point A to B.

3.2 Monetary Policy Contraction

Monetary policy contraction is when discount rate is increased or reserve ratios are increased or bonds and treasury bills are sold to the public in order to reduce money supply in circulation and slow down economic growth. Monetary policy contraction is adopted during a boom to slow down economic growth.

Figure 2: Monetary policy contraction



Source: Blanchard. O, 2007

At the same output level (Y^1) interest rate is increased from i^1 to i^2 . This leads to a reduction in money supply, LM¹ shifts upwards to the left to LM². Output reduces from Y^1 to Y^2 . The equilibrium point shifts from point A to B.

UNIT: 6

TOPIC: MONEY AND BANKING

LEARNING OBJECTIVE:

- ❑ Students must know the types of money and how money is created in the economy. The functions of the central bank and the commercial banks will also be analyzed in detail.
- ❑ At the end of the lesson, students must be conversant of the demand for money arises and how money is supplied to meet the demand.

SUB- TOPICS:

- Definition and functions of money
- Types of money
- The demand for money
- The supply for money
 - The central bank
 - The functions of the commercial banks
- Creation of money
 - Multiple deposit expansion
 - Money multiplier

1. DEFINITION AND FUNCTIONS OF MONEY

1.1 Definition of Money

- Money is any good that is widely used and accepted in transactions involving the transfer of goods and services from one person to another. Economists differentiate among three different types of money: **commodity money**, **fiat money**, and **bank money**.
- Commodity money is a good whose value serves as the value of money. Gold coins are an example of commodity money. In most countries, commodity money has been replaced with fiat money.
- Fiat money is a good, the value of which is less than the value it represents as money. Dollar bills are an example of fiat money because their value as slips of printed paper is less than their value as money.

- Bank money consists of the **book credit** that banks extend to their depositors. Transactions made using **checks** drawn on deposits held at banks involve the use of bank money.

1.2 FUNCTIONS OF MONEY

Money is used as a medium of exchange, store of value and unit of account

1.2.1 Medium of exchange

- Money's most important function is as a medium of exchange to facilitate transactions. Without money, all transactions would have to be conducted by **barter**, which involves direct exchange of one good or service for another. The difficulty with a **barter system** is that in order to obtain a particular good or service from a supplier, one has to possess a good or service of equal value, which the supplier also desires.
- In other words, in a barter system, exchange can take place *only* if there is a **double coincidence of wants** between two transacting parties. The likelihood of a double coincidence of wants, however, is small and makes the exchange of goods and services rather difficult. Money effectively eliminates the double coincidence of wants problem by serving as a medium of exchange that is accepted in all transactions, by all parties, regardless of whether they desire each others' goods and services.

1.2.2 Store of value

- In order to be a medium of exchange, money must hold its value over time; that is, it must be a store of value. If money could not be stored for some period of time and still remain valuable in exchange, it would not solve the double coincidence of wants problem and therefore would not be adopted as a medium of exchange. As a store of value, money is not unique; many other stores of value exist, such as land, works of art, and even baseball cards and stamps. Money may not even be the best store of value because it depreciates with inflation.
- However, money is more **liquid** than most other stores of value because as a medium of exchange, it is readily accepted everywhere. Furthermore, money is an easily transported store of value that is available in a number of convenient denominations.

1.2.3 Unit of account

- Money also functions as a unit of account, providing a *common measure of the value* of goods and services being exchanged. Knowing the value or price of a good, in terms of money, enables both the supplier and the purchaser of the good to make decisions about how much of the good to supply and how much of the good to purchase.

2. DEMAND FOR MONEY

The **demand for money** is affected by several factors, including the level of income, interest rates, and inflation as well as uncertainty about the future. The way in which these factors affect money demand is usually explained in terms of the three motives for demanding money: the **transactions**, the **precautionary**, and the **speculative** motives.

2.1 Transactions motive

The **transactions motive** for demanding money arises from the fact that most transactions involve an exchange of money. Because it is necessary to have money available for transactions, money will be demanded. The total number of transactions made in an economy tends to increase over time as income rises. Hence, as income or GDP rises, the **transactions demand** for money also rises.

2.1 Precautionary motive

People often demand money as a *precaution* against an uncertain future. Unexpected expenses, such as medical or car repair bills, often require *immediate payment*. The need to have money available in such situations is referred to as the **precautionary motive** for demanding money.

2.3 Speculative motive

- Money, like other stores of value, is an asset. The demand for an asset depends on both its **rate of return** and its **opportunity cost**. Typically, money holdings provide *no* rate of return and often depreciate in value due to inflation. The opportunity cost of holding money is the interest rate that can be earned by lending or investing one's money holdings. The **speculative motive** for demanding money arises in situations where holding money is perceived to be *less risky* than the alternative of lending the money or investing it in some other asset.
- For example, if a stock market crash seemed imminent, the speculative motive for demanding money would come into play; those expecting the market to crash would sell their stocks and hold

he proceeds as money. The presence of a speculative motive for demanding money is also affected by *expectations of future interest rates and inflation*. If interest rates are expected to rise, the opportunity cost of holding money will become greater, which in turn diminishes the speculative motive for demanding money. Similarly, expectations of higher inflation presage a greater depreciation in the purchasing power of money and therefore lessen the speculative motive for demanding money.

3. SUPPLY OF MONEY

- There are several definitions of the **supply of money**. *M1* is narrowest and most commonly used. It includes all **currency** (notes and coins) in circulation, all **checkable deposits** held at banks (bank money), and all **traveler's checks**. A somewhat broader measure of the supply of money is *M2*, which includes all of *M1* plus **savings** and **time deposits** held at banks. An even broader measure of the money supply is *M3*, which includes all of *M2* plus large denomination, long-term time deposits—for example, **certificates of deposit** (CDs) in amounts over \$100,000. Most discussions of the money supply, however, are in terms of the *M1* definition of the money supply.

3.1 The Central Bank and the supply of money

- A portion of each nation's money supply (*M1*) is controlled by a government agency known as the **central bank**. The central bank is unique in that it is the only bank that can issue currency. The Zambian central bank is called the **Bank of Zambia (BOZ)** .The Bank of Zambia issues all Zambian Kwacha bills, known as Federal Reserve Notes. Thus, BOZ has control over the supply of the national **currency**. BOZ also has control over the private **bank reserves** that banks entrust to the central bank. Banks hold a portion of their **required reserves** with the central bank because BOZ acts as a **clearing house** for all sorts of transactions between banks—for example, the processing of all checks.
-
- The central bank's *liabilities* therefore consist of all kwacha notes in circulation plus all private bank deposits held at the central bank as reserves On the *asset* side, BOZ owns a large amount of government debt in the form of **government bonds**. These bonds have been issued by the Treasury to pay for current and past government deficits. A simplified example of the central bank's balance sheet is provided in Table 1. Note that the central bank's total liabilities are equal to its total assets.

TABLE 1: The Balance Sheet of A central Bank (\$ values are in millions)

Assets		Liabilities	
Government bonds	\$300	Central bank Reserve notes	\$250
		Reserves of private	50
		banks	

- The central bank's control over the money supply stems from its ability to change the composition of its balance sheet. For example, the central bank may decide to purchase additional government bonds on

the open market from bondholders or private banks. This type of action is referred to as an **open market operation** by the central bank. In exchange for these government bonds, the central bank increases the reserves of private banks by the amount of the purchase.

-
- Banks, in turn, lend out their excess reserves and initiate the multiple deposit expansion process discussed above. Thus, when the central bank *buys* government bonds on the open market, it *increases* the supply of money by increasing bank reserves and inducing an expansion in the amount of deposits. Similarly, when the central bank *sells* some of its stock of government bonds to bondholders or private banks, the central bank compensates itself for the sale by reducing the reserves of private banks. The *sale* of government bonds by the central bank *reduces* the supply of money by reducing the reserves available to private banks and thereby decreasing the amount of deposit expansion that is possible.
- The central bank can also control the supply of money by its choice of the **reserve requirement**. Recall that the money multiplier is the reciprocal of the reserve requirement. If the central bank *increases* the reserve requirement, the money multiplier *decreases*, implying that deposit creation and the money supply are *reduced*. If the central bank *decreases* the reserve requirement, the money multiplier *increases*, causing both the creation of deposits and the money supply to expand further.

3.2 The commercial banks

- In order to understand the factors that determine the supply of money, one must first understand the role of the **banking** sector in the money-creation process. Banks perform two crucial functions. First, they receive funds from depositors and, in return, provide these depositors with a checkable source of funds or with interest payments. Second, they use the funds that they receive from depositors to make loans to borrowers; that is, they serve as **intermediaries** in the borrowing and lending process.
-
- When banks receive deposits, they do not keep *all* of these deposits on hand because they know that depositors will not demand all of these deposits at once. Instead, banks keep only a *fraction* of the deposits that they receive. The deposits that banks keep on hand are known as the banks' **reserves**. When depositors withdraw deposits, they are paid out of the banks' reserves. The **reserve requirement** is the *fraction* of deposits set aside for withdrawal purposes. The reserve requirement is determined by the nation's banking authority, a government agency known as the **central bank**. Deposits that banks are not required to set aside as reserves can be lent to borrowers, in the form of **loans**. Banks earn **profits** by borrowing funds from depositors at zero or low rates of interest and using these funds to make loans at higher rates of interest.
-
- A **balance sheet** for a typical bank is given in Table 2. The balance sheet summarizes the bank's **assets** and **liabilities**. Assets are valuable items that the bank owns and consist primarily of the bank's reserves and loans. Liabilities are valuable items that the bank owes to others and consist primarily of the bank's **deposit** liabilities to its depositors. In Table 2, the bank's assets (reserves and loans) total \$1

million. The bank's liabilities (deposits) total \$1 million. A banking firm's assets must always equal its liabilities.

TABLE 2: The Balance Sheet of A Typical Bank

Assets		Liabilities	
Reserves	\$100,000	Deposits	\$1,000,000
Loans	900,000		

You can infer from Table 2 that the **reserve requirement** in this example is 10%.

4. CREATION OF MONEY BY BANKS

4.1 Multiple deposit expansion

TABLE 3: Multiple Expansion of Deposits

Round	New deposits	New reserves	New loans
1	\$100,000	\$10,000	\$90,000
2	90,000	9,000	81,000
3	81,000	8,100	72,900
4	72,900	7,290	65,610
5	65,610	6,561	59,049
·	·	·	·
·	·	·	·
·	<u>+</u>	<u>+</u>	<u>+</u>
	\$1,000,000	\$100,000	\$900,000

Consider what happens when the same bank receives a \$100,000 deposit from one of its depositors. The bank is required to set aside 10% of this deposit, or \$10,000, as reserves. It then lends out its **excess reserves**—in this case, the remaining \$90,000 of the initial deposit. Suppose, for the sake of simplicity, that all borrowers redeposit their loans into the same bank. The bank thus receives \$90,000 in new deposits of which it sets \$9,000 aside as reserves and lends out all of its excess reserves. Suppose again that all borrowers redeposit their loans in the same bank, that the bank sets aside a portion of these deposits, and that the bank then lends out the remainder, which is again re-deposited in the bank and so on and so on. This repeated chain of events is summarized in Table 3.

- If one were to follow this **multiple deposit expansion** process to its completion, the end result would be that the bank's deposits would increase by \$1 million, its loans would increase by \$900,000, and its reserves would increase by \$100,000, all due to the initial deposit of \$100,000.

4.2 Money Multiplier Effect

The amount by which bank deposits expand in response to an increase in **excess reserves** is found through the use of the **money multiplier**, which is given by the formula

$$\text{money multiplier} = \frac{1}{\text{reserve requirement}}$$

-

In the example of deposit expansion found in Table 3 , the reserve requirement is 10%; so, the money multiplier in this case is $(1/.10) = 10$. The excess reserves resulting from the initial deposit of \$100,000 are \$90,000. Multiplying \$90,000 by the money multiplier, 10, yields \$900,000, which is the amount of *additional* deposits created by the banking system as the result of the initial \$100,000 deposit.

- In reality, loan recipients do not deposit all of their loan funds into a bank. More typically, they hold a fraction of their loan funds as currency. If some loan funds are held as currency, then there is a leakage of money out of the banking system. In this case, the money multiplier will still be greater than 1, but it will be *less* than the inverse of the reserve requirement.

LEARNING OBJECTIVE:

- ❑ At the end of the lesson, students must be conversant of the costs and benefits of inflation. The focus will be on the causes, consequences and cures of inflation. Students must know the policies to implement in order to control inflation and achieve the target inflation.

SUB- TOPICS:

- Definitions of inflation
- Measurement of inflation
- Causes of inflation
- Costs of inflation
- Cures of inflation

1. Definitions of inflation

This is a rise in the general level of prices in a given period of time (Samuelson. N,2006). Other economists have defined inflation as the persistent rise in relative prices over a given period of time. Inflation is a normal economic development as long as the annual percentage remains low; once the percentage rises over a pre-determined level, it is considered an inflation crisis.

- The rate of inflation is the percentage change of general prices in a given period of time.

(Samuelson. N, 2006) and is calculated as follows:

$$\frac{\text{Price level (t)} - \text{Price level (t-1)}}{\text{Price level (t -1) }} \times 100$$

Or

$$\frac{\text{CPI (t)} - \text{CPI (t-1)}}{\text{CPI (t -1) }} \times 100$$

Where t = current year

t-1 = previous year

- The government calculates the price level by constructing price indexes such as consumer price index, , GDP price index and producer price index

2. MEASUREMENT OF INFLATION

2.1 Consumer Price Index

- The consumer price index is the most widely used to measure inflation. The Consumer Price Index (CPI) is a measure of the average change of prices over time for a constant goods and services basket purchased by a representative consumer group.

The traditional method of constructing a CPI is prices for a market basket goods (common basket) such as food, clothing , shelter, transportation purchased for daily use. The index is then constructed by weighting each price according to the economic performance of the commodity.

- Simple method of calculating CPI

Assume that the market basket has two commodities: textbooks and coffee consumed in quantities of 4 and 2 respectively. The prices of these goods are as follows:

<u>Year</u>	<u>Price of Textbooks</u>	<u>Price of Coffee</u>
2000	\$1	\$2
2001	\$2	\$3
2002	\$3	\$4

Now that we know the contents of the market basket and the price of the relevant goods in the years of interest, the next step is calculate the cost of the market basket for each year.

<u>Year</u>	<u>Cost of Market Basket</u>
2000	$(4 \times \$1) + (2 \times \$2) = \$8$
2001	$(4 \times \$2) + (2 \times \$3) = \$14$
2002	$(4 \times \$3) + (2 \times \$4) = \$20$

The last step is to use the market-basket costs to calculate the CPI (assuming 2000 is the base year):

General formula : Total cost for current year divided by total cost for base year multiply by a hundred.

<u>Year</u>	<u>Consumer Price Index</u>
2000	$(\text{Cost}_{00}/\text{Cost}_{00}) \times 100 = \$8 / \$8 \times 100 = 100$
2001	$(\text{Cost}_{01}/\text{Cost}_{00}) \times 100 = \$14 / \$8 \times 100 = 175$
2002	$(\text{Cost}_{02} / \text{Cost}_{00}) \times 100 = \$20 / \$8 \times 100 = 250$

The rate of inflation for 2001 is calculated as follows:

$$\frac{175 - 100}{100} \times 100 = 75\%$$

Note that the CPI for the base year is always 100

2.2 The GDP deflator

- The deflator is used in the absence of CPI data. The deflator is the measure of the average change of prices over time not a fixed goods and services basket produced in the economy.

2.3 Producer price index

- Other various indexes have been devised to measure different aspects of inflation such as the Producer Price Index (PPI) that measures inflation at earlier stages of the production process; The CPI is generally the best measure for inflation.

3. CAUSES OF INFLATION

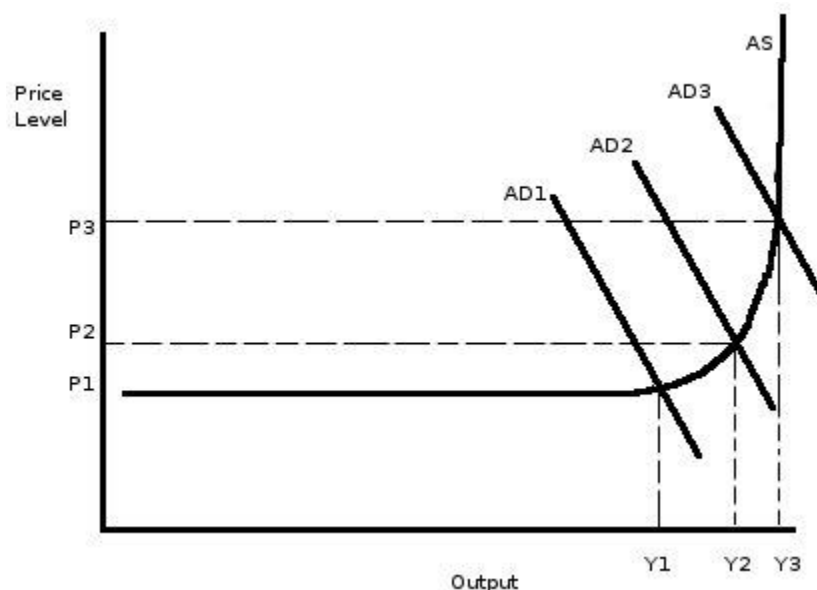
There are many causes for inflation and these depend on a number of factors.

3.1 Quantity Theory of Money

The quantity theory of money states that there is a direct relationship between the quantity of money in an economy and the level of prices of goods and services sold. According to QTM, if the amount of money in an economy doubles, price levels also double, causing inflation (the percentage rate at which the level of prices is rising in an economy). The consumer therefore pays twice as much for the same amount of a good or service. Therefore Inflation happens when governments print an excess of money to deal with a crisis. As a result, prices end up rising at an extremely high speed to keep up with the currency surplus.

3.2 Demand -Pull Inflation

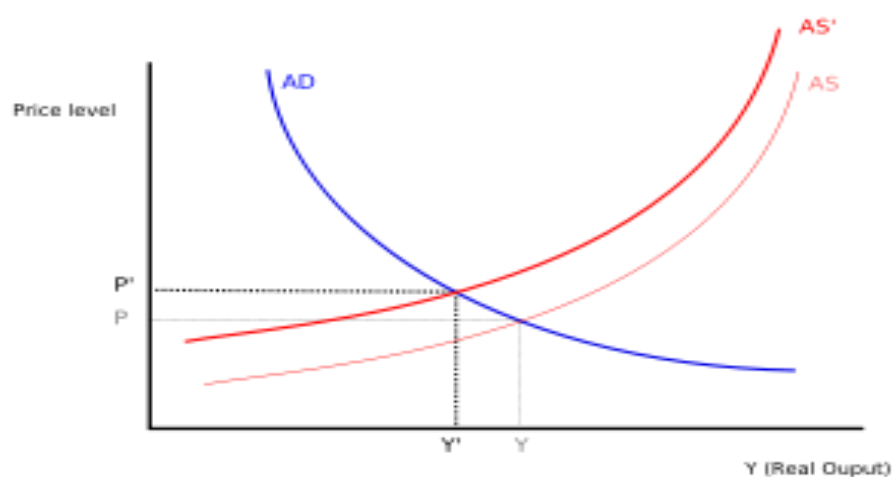
Figure 1: Demand –Pull Inflation



Demand-pull inflation happens where there is 'too much money chasing too few goods'. Excessive growth in demand literally pulls prices up. Demand-pull inflation is asserted to arise when aggregate demand in an economy outpaces aggregate supply. It involves inflation rising as real gross domestic product rises and unemployment falls. This is commonly described as "*too much money chasing too few goods*". More accurately, it should be described as involving "*too much money spent chasing too few goods*", since only money that is spent on goods and services can cause inflation. This would not be expected to persist over time due to increases in supply, unless the economy is already at a full employment level.

Figure 1 shows the demand pull inflation where as more firms will employ people, the more people are employed, and the higher aggregate demand will become. This greater demand will make firms employ more people in order to increase output. Due to capacity constraints, this increase in output will eventually become so small that the price of the good will rise. At first, unemployment will go down, shifting AD1 to AD2, which increases Y by $(Y_2 - Y_1)$. This increase in demand means more workers are needed, and then AD will be shifted from AD2 to AD3, but this time much less is produced than in the previous shift, but the price level has risen from P2 to P3, a much higher increase in price than in the previous shift. This increase in price is called inflation.

3.3 Cost - Push Inflation



Source: Saving .R. Thomas, 2006

Cost-push inflation is a type of inflation caused by substantial increases in the production costs, which leads to an increase in the price of the final product in the cost of important goods or services where no suitable alternative is available. For example, if raw materials increase in price, this leads to

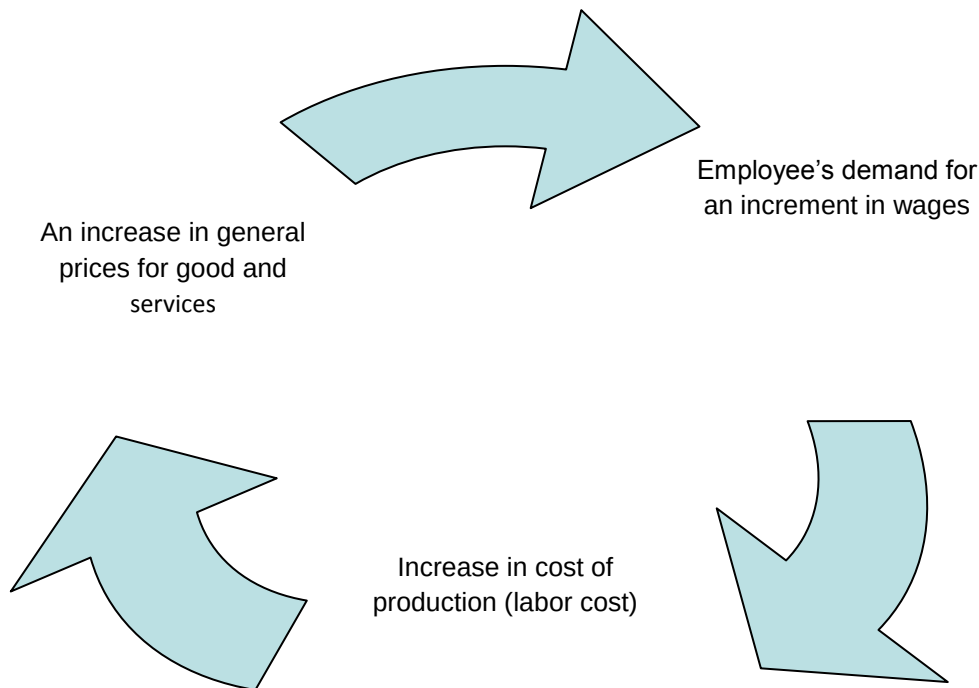
the cost of production increasing, which in turn leads to the company increasing prices to maintain steady profits.

Monetarist economists such as Milton Friedman argue against the concept of cost-push inflation because increases in the cost of goods and services do not lead to inflation without the government and its central bank cooperating in increasing the money supply. The argument is that if the money supply is constant, increases in the cost of a good or service will decrease the money available for other goods and services, and therefore the price of some those goods will fall and offset the rise in price of those goods whose prices have increased.

3.4 Wage-price spiral

- Cost-push and demand-pull inflation can interact to cause a wage-price spiral. Rising labor costs can also lead to inflation. As workers demand wage increases, companies usually chose to pass on those costs to their customers.

Figure 2; Wage - price spiral



3.5 International lending and debts

- Inflation can also be caused by international lending and national debts. As nations borrow money, they have to deal with interests, which in the end cause prices to rise as a way of keeping up with their debts. A deep drop of the exchange rate can also result in inflation, as governments will have to deal with differences in the import/export level.

3.6 Increase in fuel prices

- Finally, inflation can be caused by federal taxes put on consumer products such as fuel. As the taxes rise, suppliers often pass on the burden to the consumer; the catch, however, is that once prices have increased, they rarely go back, even if the taxes are later reduced. Wars are often cause for inflation, as governments must both recoup the money spent and repay the funds borrowed from the central bank.

A situation that has been often cited of this was the oil crisis of the 1970s, which some economists see as a major cause of the inflation experienced in the Western world in that decade. It is argued that this inflation resulted from increases in the cost of petroleum imposed by the member states of OPEC. Since petroleum is so important to industrialised economies, a large increase in its price can lead to the increase in the price of most products, raising the inflation rate. This can raise the normal or built-in inflation rate, reflecting adaptive expectations and the price/wage spiral, so that a supply shock can have persistent effects

Note that war often affects everything from international trading to labor costs to product demand, so in the end it always produces a rise in prices.

4. EFFECTS OF INFLATION

- Inflation affects different groups differently. Since the beginning of the 1980s, inflation has superseded unemployment as the principal economic target for the government. The principal reason for switching was because of the costs of inflation to the economy, and because they felt that if they could get low and stable inflation, full employment would naturally follow. The seriousness of the costs of inflation depends mainly on whether it is **anticipated** (expected) or unanticipated

4.1 Anticipated inflation

If people anticipate the inflation, then the effects will be less, as they will build these expectations into their behavior. Nevertheless, there will be costs for firms who have to keep changing their prices. These are called '**menu costs**'.

As individuals we will be less likely to hold as much cash, as it loses its value quicker when there is inflation. This means we will have to go to the bank more often to get cash out. These are called '**shoe leather costs**'.

4.2 Un –anticipated inflation

The costs are more serious if the inflation is **unanticipated**. If this happens, then we haven't allowed for it, and so wage levels get distorted. Even more serious than that is the effect on prices. Prices are fundamental to a market economy working efficiently, and if price changes become less predictable then the '**price signals**' **will not work properly** and this can lead to an inefficient allocation of resources - allocative inefficiency.

The higher the level of inflation, the more difficult it is to predict. This will almost certainly lead to higher costs. The moral of this is: low inflation is good while high inflation is bad for the economy.

5. CURES OF INFLATION

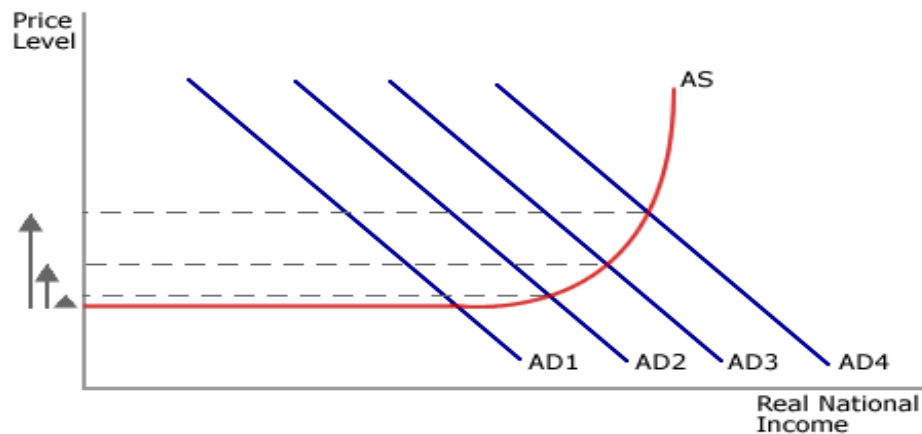
The cure has to relate to the cause. If inflation is caused by demand growing faster than the economy can cope with (**demand-pull inflation**), the remedy is to control the level of demand. If inflation is caused by a lack of capacity or by costs rising in the economy, then supply-side solutions may be required. If inflation is caused by excessive monetary growth, then it will be most appropriate to put in place policies to control the level of money supply growth. In practice, all of these are inter-linked and cannot be looked at in isolation.

5.1 Demand-side policies

If the level of demand is growing, then it will be necessary to use policies that control it. These are known as demand-side policies.

If the level of demand in the economy grows too fast, then this may cause inflation. This is shown in the diagram below. As aggregate demand shifts to the right, the price level increases - **inflation**.

Figure 1: Demand –side policies



Source: Saving .R. Thomas, 2006

Controlling demand-pull inflation entails controlling the level of demand. If demand is growing too fast then **deflationary** policies have to be put in place. Deflationary policies reduce the level of demand in the economy, and so avoid the dangers of the economy bursting .The demand-side policies are available are fiscal policy and monetary policy.

- **Fiscal policy** is the use of government expenditure and taxation to influence the economy. To reduce aggregate demand, the government reduces government spending or increase taxes in order to reduce disposable income for spending. Therefore consumption (a key component of aggregate demand) is reduced.
-
- Interest rates are the main weapon of **monetary policy**. Increasing interest rates will have a variety of effects on aggregate demand. Remember that aggregate demand includes all spending in the economy, and so is made up of:

$$\text{Aggregate Demand} = C + I + G + (X-M)$$

(where C is consumption, I is investment, G is government expenditure, X is exports and M is imports)

People who have borrowed money (including those with mortgages on their houses) will find that their loan payments have increased. This leaves them less money to spend on other things, and so reduces the level of consumption.

Firms will be affected as well. To fund their investment plans, and often their everyday spending, they borrow money. They too will face higher repayments and so may find many investment projects less profitable. This will reduce investment.

Governments are also borrowers because they have to pay interest on the National Debt. An increase in interest rates will raise the cost to them of servicing that debt. This will lead to a trade-off with other spending. The government may have to cut spending on some other public service.

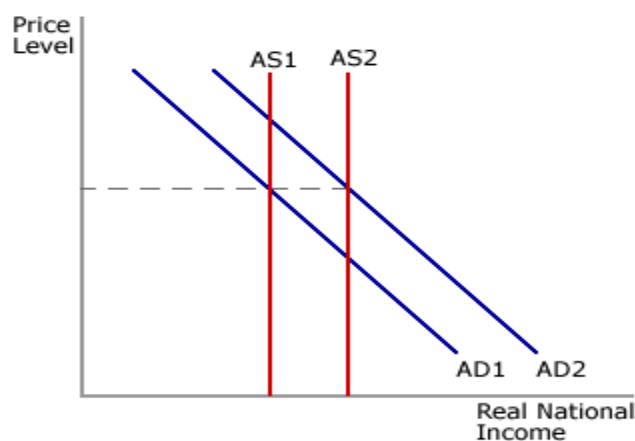
Higher interest rates may attract more overseas investment into the UK. This will lead to an increased demand for sterling and may lead to an appreciation in the exchange rate. This could adversely affect exports and lead to a further fall in aggregate demand.

5.2 Supply-side policies

- The economy can cope with increasing demand if the capacity of the economy grows fast enough. Demand-pull inflation only happens when aggregate demand grows faster than the ability of the economy to produce the goods. Therefore, if we can boost the capacity of the economy we may help to avoid inflationary problems. Supply-side policies are policies that are aimed at doing this.
- Supply-side policies are aimed at making markets work better and removing possible market imperfections. They could include:
 - Improved education and training
 - Reducing the power of trade unions to increase the flexibility of wage-setting
 - Reducing taxes to encourage investment and risk-taking
 - Reducing the level of tax to motivate people to work harder
 - Removing unnecessary regulations from markets that may hinder efficiency and innovation

The effect of these policies should be to shift the aggregate supply curve to the right. This is illustrated in the diagram below. The economy clearly has a greater capacity to produce and can cope with a higher level of aggregate demand before there is any increase in the price level.

Figure 2: Supply –side policies



Source: Saving .R. Thomas, 2006

5.3 Monetary and other targets

- Targets are set for various variables to help set policy. Recall that low and stable inflation is important to create a healthy environment for business and the economy to prosper. The low inflation is given a target in order to set policy appropriately.
- An inflation target is known as a final target. A target is set for the variable that is being controlled in this instant the target is inflation. Most countries have an inflation target. For example, The UK inflation target is 2%, Australia is 3% and Zambia 's target is between 4% - 8% in 2010.
- It is generally reckoned that monetary policy changes can take up to two years to have their full effect, and so a target also needs to be set for **future inflation** on the basis of forecasts for inflation.
- An intermediate target is a target set for a variable that affects the one that you are trying to control. In the case of inflation, you could set targets for the level of money supply growth or perhaps for the exchange rate. Both of these can act as an indicator of possible inflationary problems and help with setting policy.

5.4 Control of the money supply

The quantity theory of money suggests that control of the money supply will help in the fight against inflation. The remedy is to use inflation target and interest rates to influence money supply growth indirectly. However, there are various alternative techniques for controlling the money supply that have been suggested over the years such as monetary base control, direct controls on lending and reserve requirements.

5.4.1 Monetary base control

Banks are only able to lend to people and firms if they have enough liquid money available to meet their liabilities. This is because the more they lend, the more money is required in the system. So, they have to have more than enough cash available to handle all requests for cash at a given time. After all you wouldn't be very happy if you turned up at your cash machine and it just said 'sorry!' This liquid money is referred to as 'base money'. The theory of monetary base control is that if you control the amount of base money available, you will limit the amount the banks can lend, and therefore limit the growth of the money supply.

5.4.2 Direct controls on lending

Direct controls are regulations and requirements imposed on the banks and other financial institutions. They may take various forms, but usually set limits on the amounts the banks can lend. They are also often termed credit controls. These methods were dropped as the complexity and sophistication of the financial system increased. It would now be very difficult to use them, and there is little will to use them anyway. This is because they introduce all sorts of rigidities and reduce the amount of competition in financial markets.

5.4.3 Reserve requirements

Banks have to place a proportion of their liabilities as a reserve requirement with the central bank. These will often be non-interest bearing (pay no interest). The central bank could then use this reserve requirement to influence how much the banks could lend. If they raise the reserve requirement, then the banks would have to transfer money to the central bank and have less cash available to draw on when people wanted it. This would limit the amount they could lend as they would not be able to afford to generate new deposits. This would slow down the rate of growth of the money supply. Lowering the reserve requirement would have the opposite effect and would enable the banks to expand their deposits, increasing money supply growth.

UNIT: 8

TOPIC:

UNEMPLOYMENT

LEARNING OBJECTIVE:

- At the end of the lesson, students must be conversant of the different types of unemployment and competent explain the causes, consequences and cures of unemployment. Students must know the policies to implement in order to control high unemployment and achieve a desirable unemployment rate .

SUB- TOPICS:

- Defining 'full' employment
- Types of unemployment
- Measures of Unemployment
- Costs of unemployment
 - Economic costs of unemployment
 - Social costs of unemployment
- Inflation and unemployment (Phillips curve)
 - The original Philips curve
 - The critics of the Philips curve
 - The expected augmented Philips curve

1. DEFINING 'FULL' EMPLOYMENT

- This is when all available labor resources are being used in the most economically efficient way. Full employment embodies the highest amount of skilled and unskilled labor that could be employed within an economy at any given time. When the economy is said to be at full employment, it is at its natural rate of unemployment (Begg .D, 2005)

1.1 Natural rate of unemployment

- This is the lowest rate of unemployment that an economy can sustain over the long run. Keynesians believe that a government can lower the rate of unemployment (i.e. employ more people) if it were willing to accept a higher level of inflation (the idea behind the Phillips Curve). However, critics of this say that the effect is temporary and that unemployment would bounce back up but inflation would stay high. Thus, the natural, or equilibrium, rate is the lowest level of unemployment at which inflation remains stable. Also known as the "non-accelerating inflation rate of unemployment" (NAIRU).

1.2 Unemployment

- Unemployment is the number of people without jobs out of the labor force. It is an economic condition where an individual or individuals seeking jobs remain un-hired. The level of unemployment differs with economic conditions and other market forces. Basically there are five types of unemployment namely frictional unemployment, structural unemployment, cyclical unemployment, classical unemployment and seasonal unemployment.

2. TYPES OF UNEMPLOYMENT

2.1 Frictional Unemployment

- Frictional unemployment is the amount of unemployment that results from workers who are in between jobs, but are still in the labor force. This unemployment occurs when an individual is out of his current job and looking for another job. The time period of shifting between two jobs is known as frictional unemployment. Frictional unemployment is a temporary condition. Many economists have estimated the amount of frictional unemployment, with the number ranging from 2-7% of the labor force.
-
- **Frictional unemployment** is also a term used to describe unemployment that results from difficulties in matching *qualified* workers with new jobs. Many qualified workers seeking work are not able to find new jobs right away, usually because of a lack of complete information about new job openings. While it is likely that qualified workers will soon be matched with new jobs, these workers are considered frictionally unemployed during the time that they spend searching for their new jobs.

2.2 Structural Unemployment

- This is unemployment resulting from changes in the basic composition of the economy. These changes simultaneously open new positions for trained workers. An example of structural unemployment is the technological revolution. Computers may have eliminated jobs, but they also opened up new positions for those who have the skills to operate the computers. Structural unemployment occurs due to the structural changes within an economy.
- This type of unemployment occurs when there is a mismatch of skilled workers in the labor market. Some of the causes of the structural unemployment are geographical immobility (difficulty in moving to a new work location), occupational immobility (difficulty in learning a new skill) and technological change (introduction of new techniques and technologies that need less labor force). Structural unemployment depends on the growth rate of an economy and also on the structure of an industry.

- **Structural unemployment** results from structural changes in the economy that cause workers to lose jobs. The same structural changes also prevent these workers from obtaining new jobs. Structurally unemployed workers are *not qualified* for the new job openings that are available, mainly because they lack the education or training needed for the new jobs. Consequently, the structurally unemployed tend to be out of work for long periods of time, usually they learn the skills of the new jobs or until they decide to relocate.

2.3 Cyclical unemployment

- This is unemployment that relates to the cyclical trends in growth and production that occur within the business cycle. When business cycles are at their peak, cyclical unemployment will be low because total economic output is being maximized. When economic output falls, as measured by the gross domestic product (GDP), the business cycle is low and cyclical unemployment will rise. The aggregate demand for goods and services decrease and demand for labor decreases.
-
- Economists describe cyclical unemployment as the result of businesses not having enough demand for labor to employ all those who are looking for work. The lack of employer demand comes from a lack of spending and consumption in the overall economy.

2.4 Seasonal unemployment

- This is unemployment that occurs due to the seasonal nature of the job is known as seasonal unemployment. The industries that are affected by seasonal unemployment are hospitality industry, tourism industries, fruit picking and catering industries.

2.5 Classical unemployment

- Classical unemployment is also known as the real wage unemployment or disequilibrium unemployment. This type of unemployment occurs when trade unions and labor organization bargain for higher wages, which leads to fall in the demand for labor.

3. MEASURING UNEMPLOYMENT

- The **unemployment rate** measures the percentage of the total civilian labor force that are currently unemployed. The formula for the unemployment rate is given by

$$\text{unemployment rate} = \frac{\text{number of people unemployed}}{\text{number of people in the civilian labor force}}$$

- The civilian labor force consists of all civilians (non-military personnel), 16 years of age or older, who are willing to work and are not incarcerated. The number of people unemployed is determined according to certain criteria. In the U.S., an unemployed person is a member of the civilian labor force who is currently available for work and who has worked *less than one hour* per week for pay or profit.

- Furthermore, an unemployed worker must have been actively searching for work during the past month. Workers who are not actively searching for work, referred to as **discouraged workers**, are not considered a part of the civilian labor force and therefore are not counted among the unemployed.

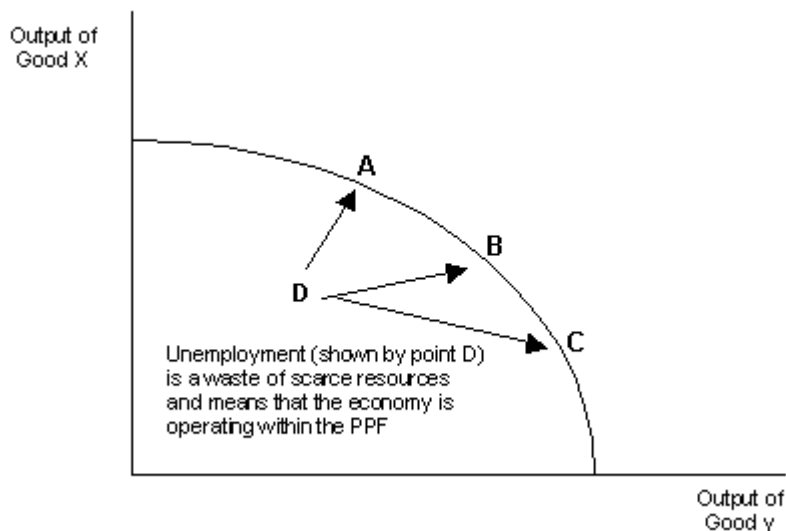
5. COSTS OF UNEMPLOYMENT

- Most economists agree that high levels of unemployment are costly not only to the individuals and families directly affected, but also to local and regional economies and the economy as a whole. We can make a distinction between the **economic costs** arising from people out of work and the **social costs** that often result.

5.1 Economic Costs

- Lost output of goods and services

Unemployment causes a **waste of scarce economic resources** and reduces the long run **growth potential** of the economy. An economy with high unemployment is producing within its **production possibility frontier**. The hours that the unemployed do not work can never be recovered.



Source; Poole. H, 2000

But if unemployment can be reduced, total national output can rise leading to an improvement in economic welfare.

- **Fiscal costs to the government**

High unemployment has an impact on government expenditure, taxation and the level of government borrowing each year. An increase in unemployment results in higher benefit payments and lower tax revenues. When individuals are unemployed, not only do they receive benefits but also pay no income tax. As they are spending less they contribute less to the government in indirect taxes.

This rise in government spending along with the fall in tax revenues may result in a higher government borrowing requirement (known as a public sector net cash requirement)

- **Deadweight loss of investment in human capital**

Unemployment wastes some of the **scarce resources** used in training workers. Furthermore, workers who are unemployed for long periods become **de-skilled** as their skills become increasingly dated in a rapidly changing job market. This reduces their chances of gaining employment in the future, which in turn increases the economic burden on government and society. See the revision page on long term unemployment

5.2 SOCIAL COSTS OF UNEMPLOYMENT

- Rising unemployment is linked to **social and economic deprivation** - there is some relationship between rising unemployment and rising crime and worsening social dislocation (increased divorce, worsening health and lower life expectancy).
-
- Areas of high unemployment will also see a decline in **real income** and **spending** together with a rising scale of relative poverty and **income inequality**. As younger workers are more geographically mobile than older employees, there is a risk that areas with above average unemployment will suffer from an ageing potential workforce - making them less attractive as investment locations for new businesses.

5.3 The duration of unemployment affects the economic and social costs

- It is clear therefore that unemployment carries substantial **economic** and **social costs**. These costs are greatest when long-term structural unemployment is high. Indeed many governments focus their labor market policies on improving the employment prospects of the long-term unemployed.

6. INFLATION AND UNEMPLOYMENT

- The Phillips Curve shows the short-term trade-off between unemployment and inflation. The original Phillips Curve has come under sustained attack – in particular from monetarist economists

6.1 Original Phillips Curve idea

- The originator of the Philips curve was AW Philips in 1958 .He plotted 95 years of data of UK wage inflation against unemployment. It seemed to suggest a short-run trade-off between unemployment and inflation. The theory behind this was fairly straightforward. Falling unemployment might cause rising inflation and a fall in inflation might only be possible by allowing unemployment to rise. If the Government wanted to reduce the unemployment rate, it could increase aggregate demand but, although this might temporarily increase employment, it could also have inflationary implications in labor and the product markets.
-
- The key to understanding this trade-off is to consider the possible inflationary effects in both labor and product markets arising from an increase in national income, output and employment:

The labor market:

As unemployment falls, some labor shortages may occur where skilled labor is in short supply. This puts extra pressure on wages to rise, and since wages are usually a high percentage of total costs, prices may rise as firms pass on these costs to their customers.

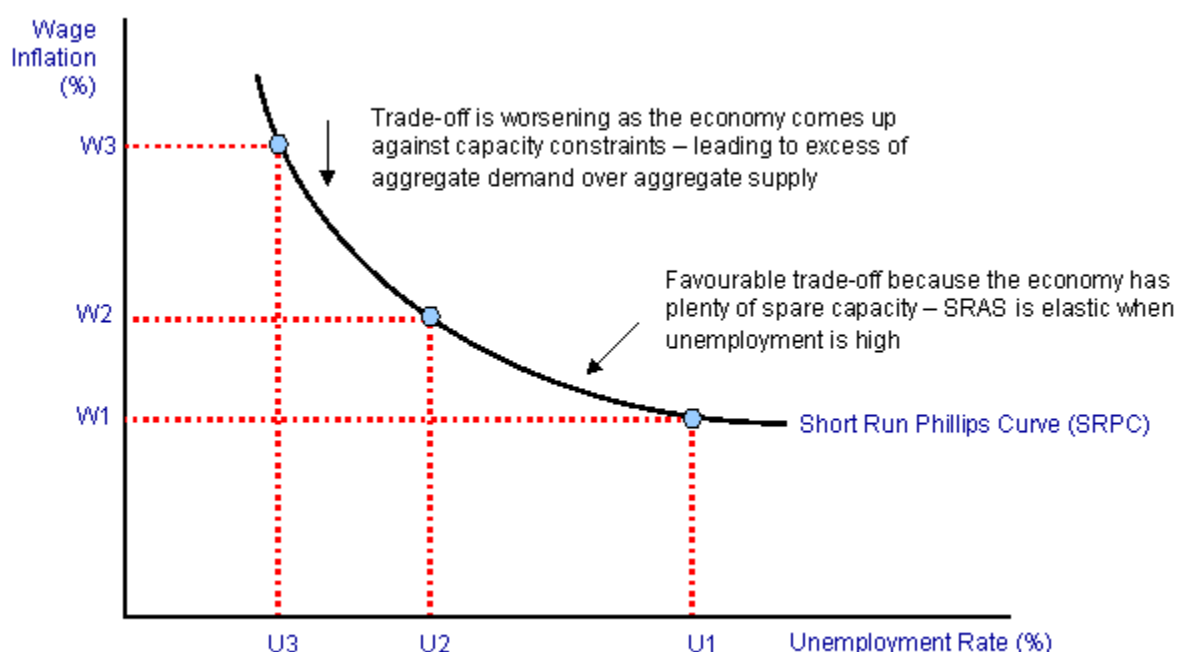
Other factor markets:

Cost-push inflation can also come from rising demand for commodities such as oil, copper and processed manufactured goods such as steel, concrete and glass. When an economy is booming, so does demand for these components and raw materials.

Product markets:

Rising demand and output puts pressure on scarce resources and can lead to suppliers raising prices to widen profit margins. The risk of rising prices is greatest when demand is out-stripping supply-capacity leading to excess demand.

Figure 1: The trade-off between unemployment and inflation



Source: Blanchard. O, 2007

6.2 Critics of the original Phillips Curve The NAIRU

- **Milton Friedman** criticized the basis for the original Phillips Curve and introduced the concept of the rate of unemployment when the rate of wage - inflation is stable. The non -accelerating inflation rate of unemployment (NAIRU) assumes that there is imperfect competition in the labor market where some workers have collective bargaining power through membership of trade unions with employers. And, some employers have a degree of monopsony power when they purchase labor inputs.
-
- According to NAIRU, the equilibrium level of unemployment is the outcome of a bargaining process between firms and workers. In this model, workers have in their minds a target real wage. This target real wage is influenced by what is happening to unemployment – it is assumed that the lower the rate of unemployment, the higher workers' wage demands will be. Employees will seek to bargain their share of a rising level of profits when the economy is enjoying a cyclical upturn.
-
- Whether or not a business can meet that target real wage during pay negotiations depends partly on what is happening to labor productivity and also the ability of the business to apply a mark-up on cost in product markets in which they operate. In highly competitive markets where there are many competing suppliers; one would expect lower mark-ups (lower profit margins) because of competition in the market. In markets dominated by monopoly suppliers, the mark-up on cost is usually much higher and potentially there is an increased share of the 'producer surpluses that workers might opt to bargain for.

- If actual unemployment falls below the NAIRU, theory suggests that the balance of power in the labor market tends to switch to employees rather than employers. The consequence can be that the economy experiences acceleration in pay settlements and the growth of average earnings. Ceteris paribus, an increase in wage inflation will cause a rise in cost-push inflationary pressure.

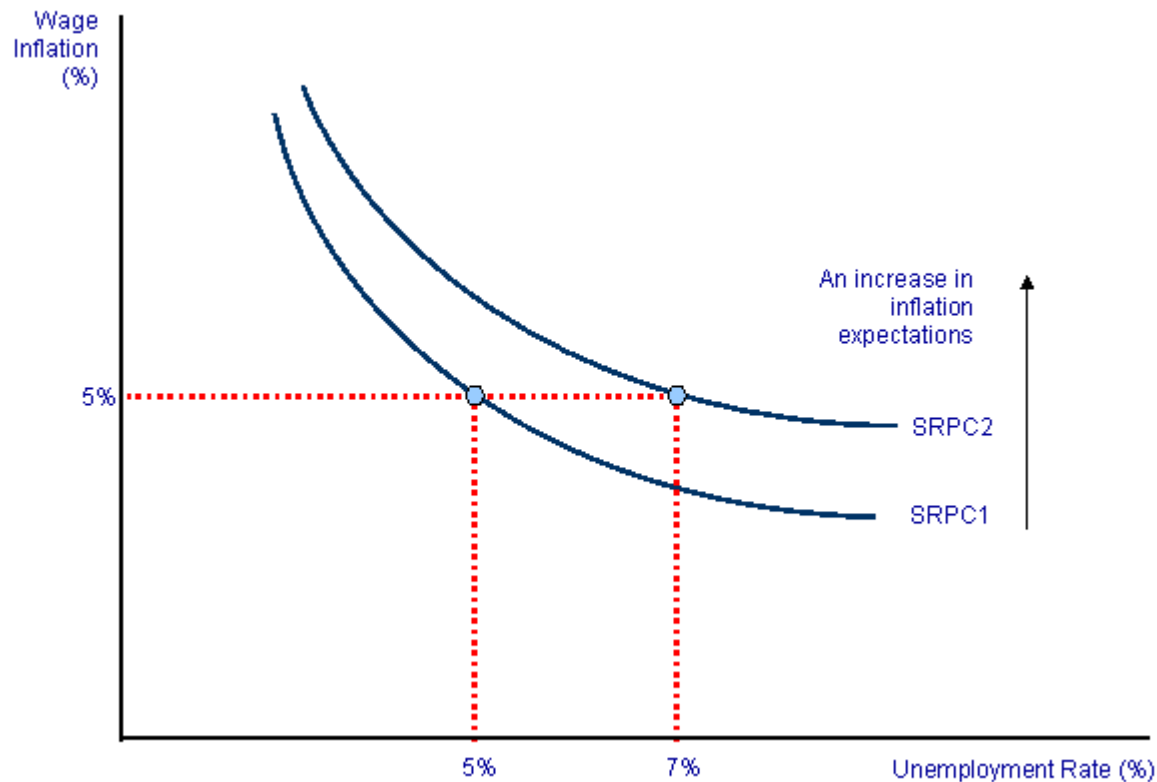
6.3 **The expectations-augmented Phillips Curve (Modern Philips Curve)**

- The original Phillips Curve idea was subjected to fierce criticism from the Monetarist school among them but that in the long run, the Phillips Curve was vertical and that there was no trade-off between unemployment and inflation.
-
- He argued that each short run Phillips Curve was drawn on the assumption of a given expected rate of inflation. So if there were an increase in inflation caused by a large monetary expansion and this had the effect of driving inflationary expectations higher, then this would cause an upward shift in the short run Phillips Curve.

Friedman introduced the idea of adaptive expectations –

This is illustrated in the next diagram – inflation expectations are higher for SPRC2. The result may be that higher unemployment is required to keep inflation at a certain target level.

Figure 2: Expected Augmented Philips Curve



Source: Blanchard. O, 2007

The **natural rate of unemployment** is the unemployment rate such that the real wage chosen in wage setting is equal to the real wage implied by price setting.

The expectations-augmented Phillips Curve argues that attempts by the government to reduce unemployment below the natural rate of unemployment by boosting aggregate demand will have little success in the long run. The effect is merely to create higher inflation and with it an increase in inflation expectations. The Monetarist school believes that inflation is best controlled through tight control of money and credit. Credible policies to keep on top of inflation can also have the beneficial effect of reducing inflation expectations – causing a downward shift in the Phillips Curve.

- The monetarist view is that attempts to boost AD to achieve faster growth and lower unemployment have only a temporary effect on jobs. Friedman argued that a government could not permanently drive unemployment down below the NAIRU – the result would be higher inflation which in turn would eventually bring about a return to higher unemployment but with inflation expectations increased along the way.

UNIT: 9

TOPIC: OPEN MARKET ECONOMY AND TRADE

LEARNING OBJECTIVE:

- ❑ At the end of the lesson, students must explain the comparative advantage theory of why nations trade, Outline arguments for free trade and protection and outline the structure of a nation's Balance of Payments.

SUB- TOPICS:

- The open Market
- International Trade
 - o Basis of trade
 - o Terms of trade
 - o Trade barriers
 - o Trade protection
- Balance of Payments
 - o Components of BOP
 - Current account
 - Capital account
 - Current account deficit
 - Zambia 's balance of payment
- Foreign Exchange rates
 - o Definition exchange rates
 - o Types of exchange rates
 - o Effects exchange rates on investments
 - o Devaluation

1. AN OPEN MARKET ECONOMY

- An open market economy is an economy that is open to trade with other nations and the government does not regulate prices and allocation of resources (Poole. H, 2000). All economic actors have an equal opportunity of entry in that market. This contrasts with a protected market in which entry is conditional on certain financial and legal requirements or which is subject to tariff barriers, taxes, levies or state subsidies which effectively prevent some economic actors from participating in them
-
- Economists then judge the "openness" of markets according to the amount of government regulation of those markets, the scope for competition, and the absence or presence of local cultural customs which get in the way of trade. In principle, a fully open market is a completely free market in which all economic actors can trade without any external constraint.

-
- In reality, few markets exist which are open to that extent, since they usually cannot operate without an enforceable legal framework for trade which guarantees security of property, the fulfillment of contractual obligations associated with transactions, and the prevention of cheating. In response to this type of criticism, the concept of open market is often redefined to mean a situation of free competition, and the inability to participate is explained as a lack of competitiveness.

2. INTERNATIONAL TRADE

- International trade allows a country to consume a combination of goods and services that exceeds its production possibilities curve. The theory of international trade assumes:
 - There are two countries
 - Each produces two goods, but initially there is no trade.

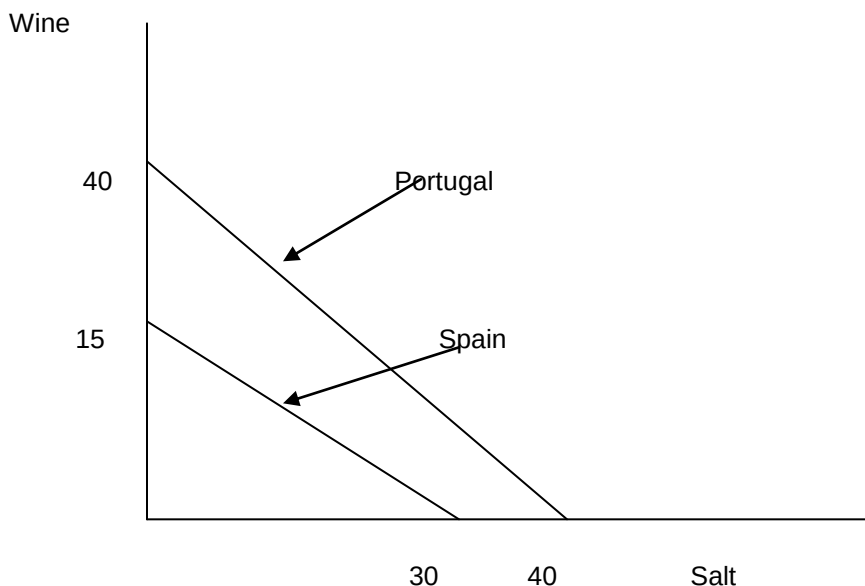
The following table shows the possible production of two goods in two economies

Table 1: Production of wine and salt

Good	Spain	Portugal
Wine	15	40
Salt	30	40

Source: Layton Robinson & Tucker, 2009

Figure 1: Pre –trade production possibility frontier



Source: Layton Robinson & Tucker, 2009

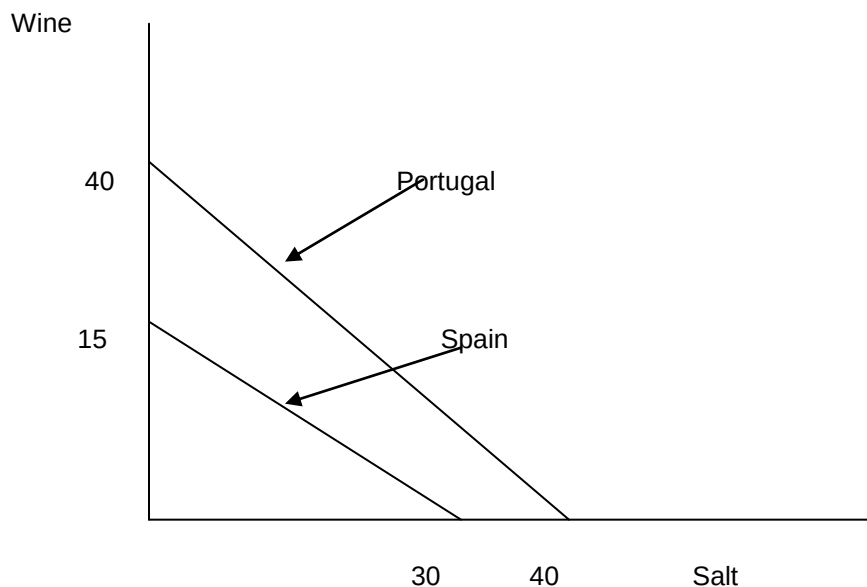
2.1 Basis of Trade

2.1.1 Adam Smith – Absolute Advantage

According to Adam Smith Model of absolute advantage in labor, the country with the relatively higher labor productivity in a commodity should completely specialize and export that commodity and import the commodity where it has relatively lower labor productivity.

Absolute advantage of the production of a commodity is represented on the diagram by a higher intercept (indicates higher labor productivity).

Figure 2: Absolute advantage production possibility frontier



Source: Layton Robinson & Tucker, 2009

Portugal has a higher intercept indicating that it has higher labor productivity in the production of wine. Based on absolute advantage, Portugal has comparative advantage in the production of wine.

2.1.2 Ricardian model – Comparative Advantage

The country with the relatively lower opportunity cost in the production of a commodity should completely specialize and export that commodity. It should, at the same time, import the commodity which it has a relatively higher opportunity cost. The country should produce the commodity in which it has the greatest comparative advantage or *countries should do what they do best*

Calculating the opportunity cost,

Table 2: Production of wine and salt in Spain & Portugal

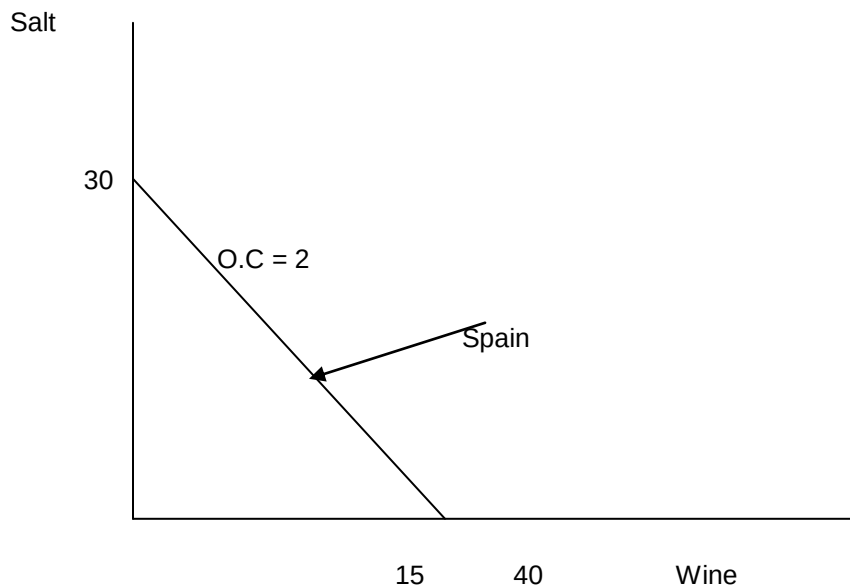
Good	Spain	Portugal
Wine	15	40
Salt	30	40

Table 3: Opportunity cost of wine in Spain & Portugal

Good	Spain	Portugal
Wine	$15/15 = 1$	$40/40 = 1$
Salt	$30/15 = 2$	$40/40 = 1$

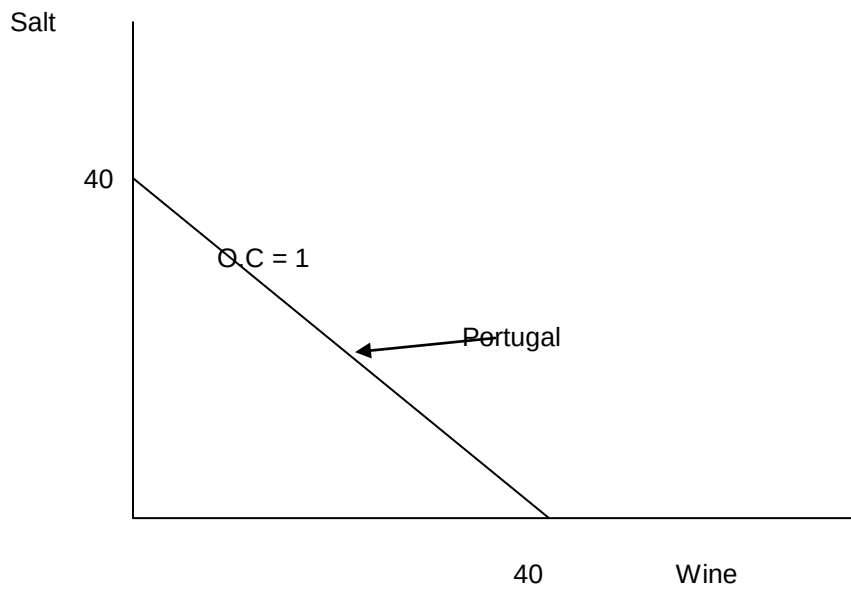
The calculations in table 3 show that Portugal has the least opportunity cost in the production of wine. This means that Portugal will only forego 1 unit of salt if it produces 1 unit of wine compared to Spain that will forego 2 units of salt if it produces 1 unit of wine. Therefore, Portugal has least opportunity cost in the production of wine and has comparative advantage in the production of wine. Portugal will specialize in the production of wine and export to Spain.

Figure 3: Opportunity cost of wine in Spain



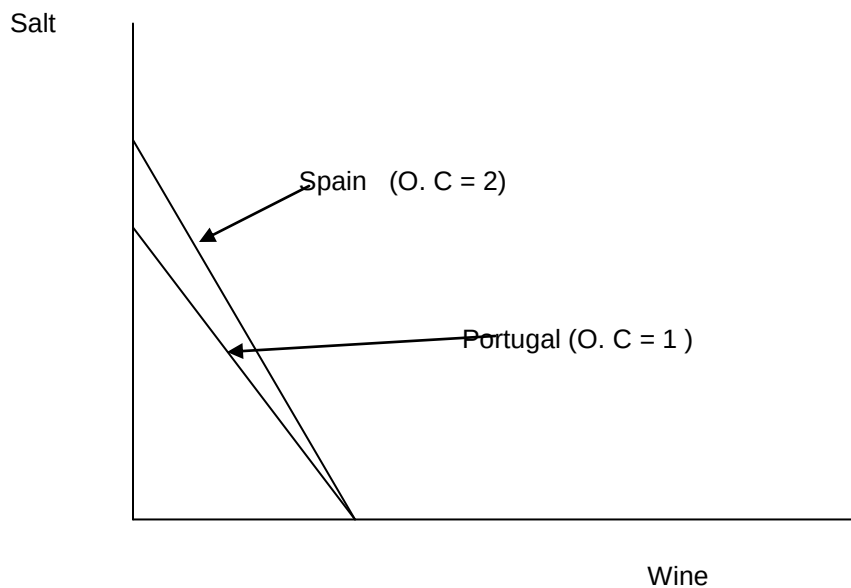
Source: Layton Robinson & Tucker, 2009

Figure 4: Opportunity cost of wine in Portugal



Source: Layton Robinson & Tucker, 2009

Figure 5: Opportunity cost of wine in Spain & Portugal



Source: Layton Robinson & Tucker, 2009

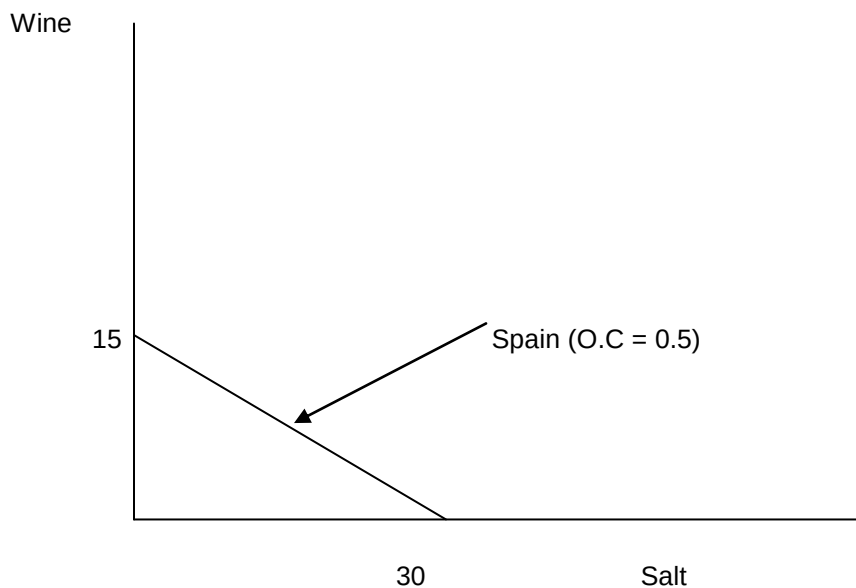
The slope of PPF represents the opportunity cost of producing wine. Since the slope of Spain's PPF is relatively steeper than that of Portugal, we can conclude that Portugal has a lower opportunity cost in the production of wine.

Table 4: Opportunity cost of Salt in Spain & Portugal

Good	Spain	Portugal
Wine	$15/30 = 0.5$	$40/40 = 1$
Salt	$30/30 = 1$	$40/40 = 1$

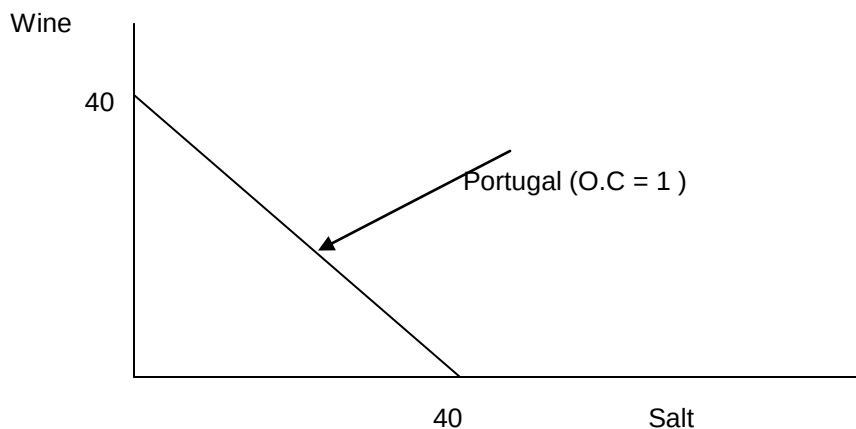
The calculations in table 4 show that Spain has least opportunity cost in the production of salt. This means that Spain will only forego 0.5 units of wine if it produces 1 unit of salt compared to Spain that will forego 1 unit of wine if it produces 1 unit of wine. Therefore, Spain has least opportunity cost in the production of salt and has comparative advantage in the production of salt. Spain will specialize in the production of salt and export to Portugal.

Figure 6: Opportunity cost of salt in Spain



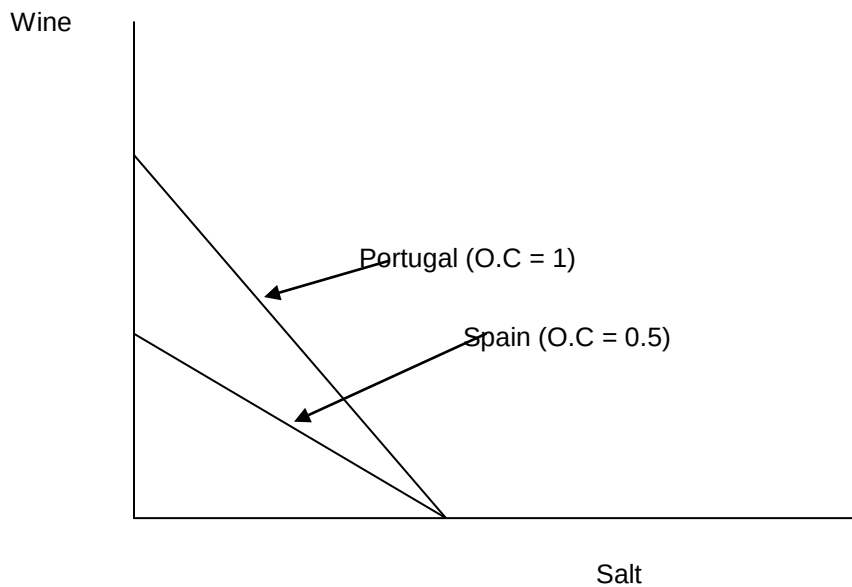
Source: Layton Robinson & Tucker, 2009

Figure 7: Opportunity cost of salt in Spain



Source: Layton Robinson & Tucker, 2009

Figure 8: Opportunity cost of salt in Spain & Portugal



Source: Layton Robinson & Tucker, 2009

The slope of PPF represents the opportunity cost of producing salt. Since the slope of Portugal's PPF is relatively steeper than that of Spain, we can conclude that Spain has a lower opportunity cost in the production of salt.

2.2 Terms of trade

- Domestic opportunity costs (pre trade prices) set the limits to the single world price which will emerge with trade. The world price is also known as international price ratio. Portugal will export wine, but only if the price is at least 1 unit of salt (before trade it could already have 1 salt for every 1 wine it gave up). Spain would be happy to pay up to 2 units of salt (Spain's pre trade cost). Any rate of exchange between the domestic price ratios will be acceptable. Note this is not dollars, but opportunity costs.

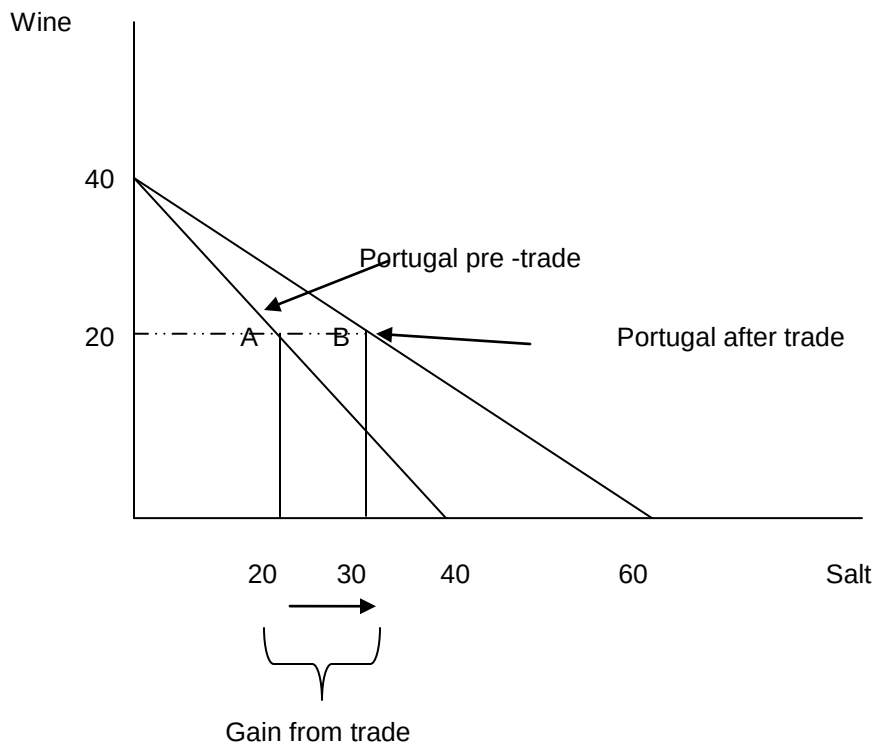
Assume the final price is 1 wine for 1.5 salt. Portugal can now get 1.5 salt for every wine it chooses to trade instead of the initial (pre –trade) 1 wine for 1 salt.

2.3 Gains from Trade

A country gains more in trade compared to producing a product locally and consuming it locally. By exporting a product where a country has least opportunity cost and importing a product where it has high opportunity cost, a country will consume more of a high opportunity cost product through trade.

Lets say Portugal chooses to export 20 wine. How many units of salt can they get now?

Figure 9: Gain from trade between Spain & Portugal



Source: Layton Robinson & Tucker, 2009

- At pre-trade point A, Portugal's consumption is 1 unit of wine for 1 unit of salt resulting into a total consumption of 20 units of wine and 20 units of salt. After trade with 1 unit of wine being traded for 1,5 units of salt, Portugal's new consumption at point B will total 20 units of wine and 30 units of salt. The gain is 10 more units of salt. Portugal is better off.

3. TRADE BARRIERS

- Trade barriers are any of a number of government-placed restrictions on trade between nations. The most common sorts of trade barriers are things like subsidies, tariffs, quotas, duties, and embargoes. The term *free trade* refers to the theoretical removal of all trade barriers, allowing for completely free and unfettered trade. In practice, however, no nation fully embraces free trade, as all nations utilize some assortment of trade barriers for their own benefit.

3.1 Types of trade barriers

a) Tariffs

- A tariff is a tax placed on imported goods. Each country has separate tariff regulations. These are a fairly common form of trade barriers, and are essentially taxes on goods as they cross the borders of a nation. Tariffs nearly always are placed on goods that are brought into the country, as opposed to goods sold as exports, although in some cases they may go both ways. Historically, tariffs were a large source of government revenue, as they could easily be collected as a tax on ships as they landed in the nation.

The five main types of tariffs include revenue, advalorem, specific, prohibitive and protective.

Revenue Tariff: A revenue tariff increases government funds. For example, countries that do not grow bananas may create a tariff on importing bananas. The government would then make money from businesses that import bananas.

An ad valorem: tariff means that the tariff applies to a percentage of the import's value such as a set number of cents on every dollar of value. A specific tariff, on the other hand, means that the tariff is not concerned with the estimated value of the imported goods, but rather is based on specific amount of the goods. A specific tariff may apply to the number of goods imported or to the weight, volume or other measurement of the goods.

Prohibitive tariff: is one that is such as high cost that it keeps the item from being imported. A protective tariff is used to raise the price of imported goods as a protective measure against the competition from foreign markets. A higher tariff allows a local company to compete with foreign competition.

Protective tariffs: can be advantageous as they can help foster the local economy, but sometimes they can also make the price of the item so expensive that companies must charge more. For example, when gas prices become too high, industries such as the trucking industry may have to charge retailers more for delivering products. The retail industry then has to mark up their items to allow for their increased transportation costs in order to make the same profit they once did. The end result is that consumers pay more for the goods.

- Tariffs, like most trade barriers, may be imposed for different reasons. Some tariffs are placed simply to earn money for the government. This might either be a flat fee on an item, or it might be based on the market value of the item. Other tariffs exist as a form of protectionism, to make imported goods more expensive than they might otherwise be, in order to protect domestic industries. For example, if a country has a fairly high wage, and high labor standards, the cost of producing a single widget might be around ten units. If a nearby country can produce a widget for three units, then imports of that country's widgets could easily drive the domestic industry out of business. So the country might place a restrictive tariff on widget imports, to make sure that domestic widgets always remained competitively priced, or even to make it unfeasible for widgets to be imported at all.

b) Subsidies

- Subsidies are another of the common trade barriers, and are often placed to protect domestic industries. Subsidies may actually be intended simply to make certain key goods affordable to citizens of the nation, but the end result can still be to make imports non-competitive. Many food crops, for example, are heavily subsidized, to ensure the citizenry has a constant supply of affordable food. Steel is also often subsidized, to ensure a nation always has a domestic steel supply, which can be crucial during times of war when normal shipping avenues may be cut off.

c) **Embargo**

- An *embargo* can be seen as the most extreme of the trade barriers. Embargoes basically prohibit the import or export of anything with another country. This is often done as a form of punishment, or to try to force the country to undergo radical change internally as a result of a weakened economic state. Historically, the embargo was used as a war tactic, and so was often considered a declaration of war. In modern times, however, although the most brutal of the trade barriers, it is usually not viewed as an act of outright aggression, although a declaration of war is often accompanied by an embargo.

d) Quotas – a limit on the quantity of a good that may be imported.

4. **TRADE PROTECTIONISM**

- Protectionism is an economic policy which is meant to benefit domestic producers of goods and services. In a nation with protectionist policies, domestic producers are insulated from competition against foreign firms by a series of barriers to import. They may also be supported directly by the government with the use of subsidies. The opposite of protectionism is free trade, in which goods are freely permitted to cross borders. Many nations support free trade, and would prefer to see protectionist economic policies barred altogether. Signatories to the General Agreement of Tariffs and Trade (GATT) and members of the World Trade Organization (WTO), for example, are typically proponents of free trade.
-
- The logic behind protectionism is that domestic industries may suffer when confronted with foreign imports which are available at cheaper prices due to lower cost of labor, more readily available natural resources, or foreign government subsidies which help the producers keep their costs low. By imposing stiff import tariffs and quotas, a government can theoretically increase the market for domestic goods, by essentially closing the market to foreign producers. This in turn is designed to benefit the domestic economy.
-
- When restrictions on imports are accompanied by government subsidies to domestic companies and government export subsidies to encourage exports of domestic products, protectionism is intended to benefit domestic companies.
-
- Supporters of protectionism argue that it can help nascent industries by insulating them from the open market until they are strong enough to function independently.
-
- Protectionism also theoretically protects domestic employment, by encouraging companies to hire domestically, and it can be used to promote living wages and better benefits for employees. Proponents point out that protectionism can also be used to pressure foreign nations to improve conditions for their workers.

Negative effects of protectionism

- Protectionism leads to lack of competition. Domestic companies may have less interest in developing innovative new products, sticking with old inventions and technologies. They may also face export barriers, because foreign countries often respond to protectionism with protectionist policies of their own.
-
- Individual citizens can also suffer under protectionism, because they may find that prices for goods and services become inflated. Without low-cost foreign competition, companies can afford to charge whatever they like for their goods and services, and this means that consumers may pay prices which are much higher than those paid by people in other regions of the world. They may also chafe at the lack of innovation, or lobby for a greater freedom to choose between products.

5. FREE TRADE BODIES

- A number of free trade bodies exist in the world to try to curtail the use of trade barriers by nations. The World Trade Organization is perhaps the widest reaching of these bodies, and it enforces strict rules against member nations, restricting the acceptable use of things like tariffs. As a result, some countries have begun using trade barriers that are not tariffs, but have similar effects. The European Union, for example, does not allow the import of many genetically-modified organisms, which effectively bans the vast majority of food imports from the United States. In recent years, groups like the WTO have begun to look at these forms of trade barriers as well, and to strip them when possible.

5.1 World trade Organization

- According to the World Trade Organization (WTO), its goal is to help producers of goods and services, exporters, and importers conduct their business. This international organization accomplishes this mainly by helping countries establish and agree upon trade rules. Without such an organization, it is likely that trade relations between foreign entities could be difficult, unfair, and inconsistent.
-
- The World Trade Organization was created in 1995 and has three official languages: French, English and Spanish. The organization has hundreds of members around the globe, however, and aims to work in the best interest of all of them. Most decisions made by the WTO are matters of consensus. This means that all members agree upon them.
-
- A trade agreement usually contains both the rights and obligations of the agreeing parties. WTO agreements can generally be divided into two categories. Multilateral agreements refer to those that all World Trade Organization members have agreed to. Plurilateral agreements are those that only some members have agreed to.
-
- Different countries tend to have different cultures. Many people in business realize that culture can be a major factor in determining how affairs are conducted. Other factors such as previous experiences, both good and bad, can determine the business decisions that are made. These are just two of many

examples of factors that could cause trade barriers or trade inconsistencies if such business arrangements were left to the whim of individuals.

-
- One thing that the World Trade Organization aims to do is normalize trade. This means that certain practices should be kept consistent. For example, the organization may secure agreements among trade partners that prevent politics from playing a role in the initiation or rise of tariffs.
-
- In addition to easing trade transactions, the World Trade Organization also aims to eliminate discrimination from international exchange. This can produce the dual benefits of allowing producers access to foreign markets and allowing consumers access to foreign goods. This can also prevent substantial economic growth for some while economic growth is unfairly or unethically inhibited for others.

5.2 EUROPEAN UNION

- The European Union, or EU, is a collaborative effort between 27 European countries to form a mutually beneficial economic and policy community. Since 1993, the EU has worked to increase economies and spread human rights advances worldwide. The goals of the European Union include uniting Europe toward common goals and providing aid to developing nations.
-
- After World War II, Europe was a fractured area, divided by political and cultural differences. Several attempts to promote a regional governing body met with varying success, including the European Coal and Steel Community and the European Community. Because of the power ideological and political differences between Western and Eastern Europe, a true community could not be formed until after the end of the Cold War
-
- In 1992, the Maastricht Treaty was signed by member nations, bringing the European Union into effect. The treaty outlined three pillars of the union: European Communities, Common Foreign and Security Policy, and Justice and Home Affairs. The treaty also made provisions to admit many of the nations of Eastern Europe. In 2001, the Treaty of Nice further expanded provisions for new nations.
-
- To become a member of the European Union, a nation must be in conformance with a series of standards called the Copenhagen criteria. These detail geographic and political necessities for member nations. Included standards dictate human rights laws, democracy, protective law for minorities, and a market economy. As of 2008, three states are candidates undergoing review: Turkey, Croatia, and the Republic of Macedonia. Several other nations are identified as potential candidates for future admission, including Albania and Serbia.
-
- One of the greatest achievements of the European Union is the establishment of a single market economy. Between member nations, trade is largely unrestricted. While nations maintain separate laws

on taxation and trade standards, EU members agree to basic trade laws between their countries. Almost all products created by one nation are legal for trade in all of the other countries.

-
- Most nations belonging to the EU have adopted a common currency, called the Euro. The Euro is overseen by the European Central Bank, in an effort to promote all economies that incorporate the use of the currency. As of 2008, 15 nations use the Euro, collectively called the Euro zone. Other EU members must meet specific financial and economic standards before being allowed to adopt the currency. Slovenia was the first of the countries from the 2004 expansion to meet euro criteria.

.6. THE BALANCE OF PAYMENTS

The balance of payments is a summary of international transactions between one country and others over a period of time. This summary records the nature and value of inflows and outflows of goods, services and financial assets. The balance of payments comprises of the current account and the capital account.

6.1 The current account

The current account records trade in goods and services. It consists of balances on

- merchandise trade (goods) that is exports less imports
- services (e.g. tourism, education)
- income (e.g. dividends and interest payments)
- transfers.

6.2 The capital account

The capital account records dollar payments flows of purchases of foreign assets by Australians, and of domestic assets by foreigners. The assets referred to are 'investment assets such as bonds, securities, property, shares

Note that any income flowing from the ownership of these assets is recorded as income in the current account.

Current and capital account offset each other. The current and capital accounts should balance (any difference is a measurement error). This is because any deficit on the current account is financed by a surplus on the capital account – foreigners accommodate additional current expenditure by becoming investors in the country.

6.3 The current account deficit

- Most countries have experienced a current account deficit (CAD) in every year since 1960. This means that countries spend more than the income they generate, so foreigners are accumulating domestic assets.
-
- The CAD is not likely to prove a problem provided overseas funds are used to finance investment goods which earn income in the future.
-
- There has been a significant rise in foreign liabilities and debt over the past 25 years in most developing countries. Much of this is explained by financial deregulation and the freeing up of international financial markets over the last 20 years in direct investment, portfolio investment
-
- It is important to note that the amount of foreign liabilities is not a concern provided the debt can be serviced (just like a housing loan). This is more likely if liabilities have been incurred to generate income.

6.4 Zambia's balance of payment

Table 1: 2000 Zambia's balance of payments in millions of US dollars.

ble 1: 2000 Zambia's balance of payments in millions of US dollars.

<u>Current Account</u>	-584
<i>Balance on goods (exports – imports)</i>	-221
<i>Balance on services (exports – imports)</i>	-226
<i>Balance on income (income inflow – income outflow)</i>	-120
<i>Current transfers</i>	-18
<u>Capital Account</u>	153
<u>Financial Account</u>	-274
<i>Direct investment abroad</i>	...
<i>Direct investment in Zambia</i>	122
<i>Portfolio investment assets</i>	...
<i>Portfolio investment liabilities</i>	-1
<i>Other investment assets</i>	-85
<i>Other investment liabilities</i>	-309
<u>Net Errors and Omissions</u>	185
<u>Reserves and Related Items</u>	520

Source: IMF, 2000

Zambian trade is normally in rough balance. However, a heavy debt burden gives the country a current account deficit, and hard currency is often in short supply. Total debt service payments in 1997 equaled \$277 million, or about 21% of export earnings. In the early 2000s, the trade deficit worsened due to mining related imports needed to reform the privatized copper industry. Nonetheless, an improvement in official and commercial inflows, supported by a resumption of concessional donor support, was expected to prompt a recovery.

The US Central Intelligence Agency (CIA) reports that in 2001 the purchasing power parity of Zambia's exports was \$876 million while imports totaled \$12.05 billion resulting in a trade deficit of \$11.174 billion.

The International Monetary Fund (IMF) reports that in 2000 Zambia had exports of goods totaling \$757 million and imports totaling \$978 million. The services credit totaled \$114 million and debit \$340 million.

7. EXCHANGE RATES

- The exchange rate is the price of one currency expressed in terms of another currency. It is also defined as the price of the domestic currency for one unit of a foreign currency.

For example if ZMK 5000 buys \$1 dollars, the exchange rate is expressed as:

$$\frac{\text{ZMK 5000}}{\$1} = 5000.$$

7.1 TYPES OF EXCHANGE RATES

a) Nominal exchange rate

The nominal exchange rate ϵ is defined as the number of units of the domestic currency that can purchase a unit of a given foreign currency.

$$\epsilon = \frac{\text{ZMK 5000}}{\$1} = 5000.$$

Where :ZMK 5000 is also referred to as the price currency or term currency

: \$1 is also referred to as the price commodity currency or base currency

A decrease in this variable is termed nominal appreciation of the currency. (Under the fixed exchange rate regime, a downward adjustment of the rate ϵ is termed revaluation.) An increase in this variable is termed nominal depreciation of the currency. (Under the fixed exchange rate regime, an upward adjustment of the rate ϵ is called devaluation.)

Nominal exchange rates are established on currency financial markets called "**forex** markets", which are similar to stock exchange markets. Rates are usually established in continuous quotation, with newspaper reporting daily quotation (as average or finishing quotation in the trade day on a specific market). Central bank may also fix the nominal exchange rate. In most parts of the world, the market convention that determines which is the base currency and which is the term currency is in the order : EUR – GBP – AUD – NZD – USD – others.

Quotes using a country's home currency as the price currency (e.g., EUR 0.735342 = USD 1.00 in the euro zone) are known as direct quotation or price quotation (from that country's perspective) and are used by most countries. Quotes using a country's home currency as the unit currency (e.g., EUR 1.00 = USD 1.35991 in the euro zone) are known as indirect quotation or quantity quotation and are used in British newspapers and are also common in Australia, New Zealand and the euro zone.

b) Real exchange rates

This is exchange rates that have been adjusted for the inflation differential between two countries. While the nominal exchange rate shows the number of domestic currency one receives for a foreign currency (or vice versa), it does not show the purchasing power of the domestic currency versus that of a foreign currency. It is customary to distinguish nominal exchange rates from real exchange rates.

While two currencies may have a certain exchange rate on the foreign exchange market, this does not mean that goods and services purchased with one currency cost the equivalent amounts in another currency. This is due to different inflation rates with different currencies. Real exchange rates are thus calculated as a nominal exchange rate adjusted for the different rates of inflation between the two currencies.

The is denoted as

$$Q = \epsilon \times P^* / P$$

Where ϵ is the nominal exchange rate

P^* : prices for foreign goods

P : prices for domestic goods

This rate tells us how many times more goods and services can be purchased abroad (after conversion into a foreign currency) than in the domestic market for a given amount. In practice, changes of the real exchange rate rather than its absolute level are important. An increase in the real exchange rate Q is termed appreciation of the real exchange rate, a decrease is termed depreciation. In contrast to the nominal exchange rate, the real exchange rate is always 'floating', since even in the regime of a fixed nominal exchange rate ϵ , Q can move via price-level changes.

Note that the Real appreciation means an increase in the real exchange rate Q , whereas nominal appreciation means a decrease in the nominal exchange rate ϵ . Rather than focusing on the nominal exchange rate, it is more sensible to consider the real exchange rate when assessing the effect of exchange rates on international trade.

Example

Assume that the domestic price level rises by 10 %, the foreign price is 5% and the nominal exchange rate is 2.50 ,then the real exchange rate is: $Q = 2.50 \times 0.02 / 0.10 = 0.5$

c) **Effective exchange rate (EXR)**

The effect exchange rate, also referred to as the multilateral rate measures the overall nominal value of a currency in the foreign exchange market. It is computed by formulating a weighted average (reflecting the importance of each country's currency in international trade) of selected bilateral rates. Several institutions such as International Monetary Fund (IMF), US Federal Reserve bank, and Morgan Guaranty Trust Co., compute and publish EXRs based on their own weighting formulas. EXR is an important measure of the overall demand and supply of a currency. Also called trade weighted exchange rate.

Effective exchange rates are computed in order to judge the general dynamics of a country's currency toward the rest of the world. One takes **a basket of different currencies**, select a (more or less) meaningful set of relative **weights**, then computes the "**effective**" exchange rate of that country's currency.

Example:

Given that the SAR 1 has a basket made up of 40% USA dollars (\$) , 30% Euros (E) , 20% Namibian dollar (N\$) and 10% Botswana pula (P) , and the nominal exchange rates for the respective currencies are:

$$\$/\text{SAR } 1 = 0.6$$

$$\text{E}/\text{SAR } 1 = 0.4$$

$$\text{N } \$/\text{SAR } 1 = 1$$

$$\text{P}/\text{SAR } 1 = 1.2$$

The effective exchange rate is calculated as:

<u>Nominal exchange rate</u>	<u>Percentages in the basket</u>	<u>EXR</u>
$\$/\text{SAR } 1 = 0.6$	40%	$0.6 \times 40\% = 0.24$
$\text{E}/\text{SAR } 1 = 0.4$	30%	$0.4 \times 30\% = 0.12$
$\text{N } \$/\text{SAR } 1 = 1$	20%	$1 \times 20\% = 0.20$
$\text{P}/\text{SAR } 1 = 1.2$	10%	$1.2 \times 10\% = 0.12$
Total		0.68

d) Real Effective Exchange Rate (REER)

This is the weighted average (effective exchange rate) of a country's currency relative to an index or basket of other major currencies adjusted for the effects of inflation. The weights are determined by comparing the relative trade balances, in terms of one country's currency, with each other country within the index. All currencies within the said index are the major currencies being traded today: U.S. dollar, Japanese yen, euro, etc

This is also the value that an individual consumer will pay for an imported good at the consumer level. This price will include any tariffs and transactions costs associated with importing the good.

This exchange rate is used to determine an individual country's currency value relative to the other major currencies in the index, as adjusted for the effects of inflation..

8. EXCHANGE RATES REGIME

- There are some basic exchange rate regimes that are used nowadays – the floating exchange rate, the pegged float exchange rate and the fixed or pegged exchange rate.

a) Floating exchange rate

This is an exchange rate in which the value of the currency is determined by the free market. That is, a currency has a floating exchange rate when its value changes constantly depending on the supply and demand for that currency, as well as the amount of the currency held in foreign reserves..

- The currency is not constrained by central bank intervention and does not have to maintain its relationship with another currency in a narrow band. The currency value is determined by trading in the foreign exchange market
- The floating exchange rate regime is also known as a dirty float or a managed float. This is because the governments always step in to address any excesses in the changes of value. The floating rates are extensively used in most countries of the world.
- An advantage to a floating exchange rate is the fact that it tends to be more economically efficient. However, floating exchange rates tend to be more volatile, depending on the particular currency.

b) **Pegged floating exchange rates**

- There are three types of pegged floats – the crawling bands, pegging with horizontal bands and crawling bands. In case of the crawling bands the rate is permitted to fluctuate within a particular band or limit and the movements are based on a particular central value. This central value is adjusted at definite periods. The entire exercise is performed in a controlled manner.
-
- In case of the crawling pegs the rates of exchange stay fixed. In case the rates are pegged with horizontal bands the rate would be allowed to move within a specified limit or band, which is 1% more than the band.

c) **Fixed or pegged exchange rates**

- This is an exchange rate that is set by the governments involved rather than being allowed to fluctuate freely with market forces. In order to keep currencies trading at the prescribed levels, government monetary authorities actively enter the currency markets to buy and sell according to variations in supply and demand.
- A country ties the value of its currency to another country's currency, gold (or another commodity), or a basket of currencies. The fixed exchange rate does not change according to market conditions therefore it is also called a pegged exchange rate.
- Pegged exchange rates are generally more stable, but, since they are set by government fiat, they may take political rather than economic conditions into account. For example, some countries peg their exchange rates artificially low with respect to a major trading partner to make their exports to that partner artificially cheap.

7.2 **Effects of currency exchange rates on investment**

- When the exchange rate between the foreign currency of an international investment and the kwacha changes, it can increase or reduce investment return. Because foreign companies trade and pay dividends in the currency of their local market, there is need to convert the cash received from dividends or the sale of the investment into kwacha. Therefore, if the exchange rate changes significantly between the time one buys and the time one sell, it can sometimes turn a positive return in the investment itself into a loss for the investment in total, or vice versa.
-
- International investment returns increase when the kwacha weakens in value against another currency, because each unit of foreign currency translates into more kwacha notes. On the other hand, if the kwacha strengthens against the foreign currency, it translates each foreign currency unit into fewer kwacha and therefore diminishes one's returns.
-

7.3 **Devaluation**

- Devaluation is a deliberate decision by a government or central bank to reduce the value of its own currency in relation to the currencies of other countries. Devaluation occurs exclusively in fixed currencies, when the currency in question is pegged to another currency.
- Governments often opt for devaluation when there is a large current account deficit, which may occur when a country is importing far more than it is exporting. Devaluation is generally undertaken by a government in order to make its country's products more competitive in world markets .When a nation devalues its currency, the goods it imports and the overseas debts it must repay become more expensive. But its exports become less expensive for overseas buyers. These competitive prices often stimulate higher sales and help to reduce the deficit.
-
- Additionally, devaluation can significantly reduce the value of investments held by foreign investors in the devaluing country

TOPIC: ECONOMIC GROWTH AND BUSINESS CYCLES

LEARNING OBJECTIVE:

- ❑ At the end of the lesson, students must know what leads to growth in the long run. This will be analyzed using different growth theories such as classical theory, neo-classical theory and endogenous theory..

SUB- TOPICS:

- Business cycles
- Definition of Growth
 - o Classical growth theory
 - o Neo –classical growth theory
 - o Endogenous growth theory
- Factors of growth

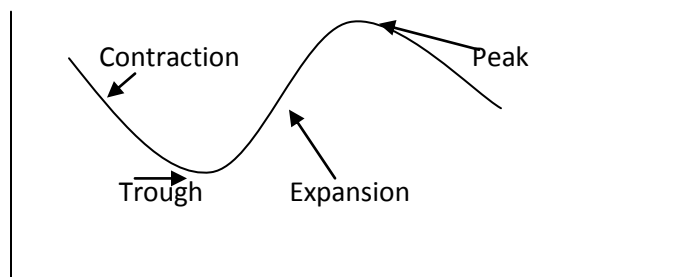
1. BUSINESS CYCLES

A business cycle is the periods of growth and decline in an economy. A business cycle is not a regular, predictable, or repeating phenomenon like the swing of the pendulum of a clock. Its timing is random and, to a large degree unpredictable.

There are four stages in the business cycle:

- i) **Contraction** - When the economy starts slowing down. There is a slowdown in the pace of economic activity.
- ii) **Trough** - When the economy hits bottom, usually in a recession. This is the lower turning point of a business cycle, where a contraction turns into an expansion
- iii) **Expansion** - When the economy starts growing again. This is when there is speedup in the pace of economic activity.
- iv) **Peak** - When the economy is in a state of "irrational exuberance." This is the upper turning of a business cycle.

Figure 1: Business cycle stages



2. DEFINITIONS OF GROWTH

Generally economic growth is an increase in the nation's capacity to produce goods and services and usually refers to real GDP growth. The increases must occur after adjustments for inflation have been made. There are different growth theories that define economic growth such as classical growth theory, neo-classical growth theory, endogenous growth theory.

2.1 Classical growth theory

The modern conception of economic growth was began by the Scottish Enlightenment thinkers such as David Hume and Adam Smith, and the foundation of the discipline of modern political economy. The theory of the physiocrats was that productive capacity, itself, allowed for growth, and the improving and increasing capital to allow that capacity was "the wealth of nations". Whereas they stressed the importance of agriculture and saw urban industry as "sterile", Smith extended the notion that manufacturing was central to the entire economy.

David Ricardo argued that trade was a benefit to a country, because if one could buy a good more cheaply from abroad, it meant that there was more profitable work to be done here. This theory of "comparative advantage" would be the central basis for arguments in favor of free trade as an essential component of growth.

2.2 Neo – classical growth theory

This model, developed by Robert Solow and Trevor Swan in the 1950s, was the first attempt to model long-run growth analytically. According to this view, the role of technological change became crucial, even more important than the accumulation of capital. This model assumes that countries use their resources efficiently and that there are diminishing returns to capital and labor increases.

The neo-classical model makes three important predictions:

- First, increasing capital relative to labor creates economic growth, since people can be more productive given more capital.
- Second, poor countries with less capital per person will grow faster because each investment in capital will produce a higher return than rich countries with ample capital.
- Third, because of diminishing returns to capital, economies will eventually reach a point at which no new increase in capital will create economic growth. This point is called a "steady state".

The model also notes that countries can overcome this steady state and continue growing by inventing new technology. In the long run, output per capita depends on the rate of saving, but the rate of output growth should be equal for any saving rate. In this model, the process by which countries continue growing despite the diminishing returns is "exogenous" and represents the creation of new technology that allows production with fewer resources. Technology improves, the steady state level of capital increases, and the country invests and grows. The data does not support some of this model's

predictions, in particular, that all countries grow at the same rate in the long run, or that poorer countries should grow faster until they reach their steady state. Also, the data suggests the world has slowly increased its rate of growth.

However modern economic research shows that the baseline version of the neoclassical model of economic growth is not supported by the evidence. Calculations made by Solow claimed that the majority of economic growth was due to technological progress rather than inputs of capital and labor.

2.3 **Endogenous growth theory**

This theory is also known as the new growth theory that was initiated by Paul Romer in the late 1980s and early 1990s. Other important new growth theorists include Robert E. Lucas and Robert J. Barro.

Unsatisfied with Solow's explanation, economists worked to "endogenize" technology in the 1980s. They developed the endogenous growth theory that includes a mathematical explanation of technological advancement. This model also incorporated a new concept of human capital, the skills and knowledge that make workers productive. Unlike physical capital, human capital has increasing rates of return. Therefore, overall there are constant returns to capital, and economies never reach a steady state.

The new growth theorists concluded that growth does not slow as capital accumulates, but the rate of growth depends on the types of capital a country invests in. Research done in this area has focused on what increases human capital (e.g. education) or technological change (e.g. innovation).

Recent empirical analyses suggest that differences in cognitive abilities, related to schooling and other factors, can largely explain variations in growth rates across countries. Cognitive abilities comprise intelligence and knowledge and are more important than education itself. Cognitive abilities are more relevant than the classical growth factor "economic freedom". In comparison of low, mean and high ability groups within societies the competence level of the high ability group is the most important, stimulating through research and innovation economic growth and other favorable aspects of countries like democracy.

3. **FACTORS OF GROWTH**

One way that output can increase is if there is an expansion in the inputs used to produce it, that is capital and labor.

There are five kinds of capital. Human-produced capital is called manufactured capital to distinguish it from the other kinds of capital. Land and natural resources are natural capital, and all the skills and knowledge possessed by humans are also a kind of capital – human capital. We also note the importance of social and financial capital, which both refer to institutional arrangements that make production possible.

Economists sometimes think about output as being generated according to a "production function," which is a mathematical relation between various inputs and the level of output.

$$\text{Output} = f(L, K)$$

In the most general sense we might say that the output of an economy should be expressed as a function of flows from all the different types of capital that make production possible. The inputs to the production function are commonly referred to as factors of production. In the production functions most commonly used by economists, the factors that are emphasized are manufactured capital and labor. Sometimes, but not always, natural resources also are included.

3.1 **Modeling Economic Growth –Solow Robert**

One very influential, and more specific, model of economic growth was developed by economist Robert Solow in 1957. In his model, he assumed that an economy-wide production function can be written in the simple form:

$$Y = A K^{0.3} L^{0.7}$$

Where:

Y is aggregate output,

A is a number based on the current state of technology as described below,

K is a quantitative measure of the size of the stock of manufactured capital,

L is the quantity of labor used during the period of time.

K and L are the only factors of production explicitly included in the model. Both capital and labor are needed for the production of output, with the exponents in the equation reflecting their relative contributions.

A is called total factor productivity, and includes all contributions to total production not already reflected in levels of K and L. Often, "total factor productivity" has been interpreted as reflecting the way in which technological innovation allows capital and labor to be used in more effective and valuable ways. For example, the development of computer word-processing greatly increased efficiency compared to the use of typewriters. Typewriters, which seem antique to us today, were themselves a huge productive advance over clerical work using pen and paper. This process of improved technological methods has resulted in an increase in labor productivity – more output can now be produced with fewer labor-hours.

After some mathematical manipulations, the production function above can be converted to an equation for the growth rate of output per worker as a function of “total factor productivity” and the growth rate of manufactured capital per worker:

Growth rate of output per worker = growth rate of total factor productivity + 0.3 (growth rate of manufactured capital per worker)

For example:

if “total factor productivity” grows at 1% per year

and capital per worker grows at 2% per year,

this equation says that output per worker will grow at 1.6% per year

$$(1\% + (0.3)2\% = 1.6\%).$$

This became known as the “growth accounting” equation.

Note that output per worker is what is commonly referred to as “labor productivity”. While labor productivity and GDP per capita are not quite equivalent (some people in the population do not work, for example), they are obviously closely related. Thus, this model implies that the way to raise income per capita—to achieve economic growth—is to increase the amount of capital that each person works with (the second term) and improve technology (the first term).

The use of the Solow growth model served to highlight some important factors in economic growth. In particular, the model led to much discussion of the role of savings in providing the basis for growing levels of manufactured capital per worker. Technological change also received attention, since this was thought to be the main driver behind growth in the value of “A.” For many years, economists tended to treat growth as primarily a matter of encouraging savings, investment, and the creation and dissemination of technology.

In more recent years, other economists have suggested that perhaps this model has directed too much attention to savings and technology. Some have argued that other factors such as good institutions that support markets, innovations in the organization of work, or access to global markets should be thought of as equally important in promoting economic growth. It is not helpful, they suggest, to fold all issues of social, human, financial and natural capital into just one, rather vague, “A” term

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